Senior Design Project2

Final Report

Intelligent Real-Time Arabic Sign Language Translation

Osama Almolike

Mohammed Badwai

Mohammed Badr

A report submitted in part fulfilment of the degree of

Bachelor's in electrical engineering

Advisor: Prof. Adel Abdennour



Department of Electrical Engineering

Faculty of Engineering, Islamic University of Madinah

G:2025/H:1447

Declaration

This report was prepared based on our work. Where other published and unpublished source materials have been used, these have been acknowledged.

Students' Names and Signatures:

Student Names	Signatures
Osama Almolike	
Mohammed Badwai	
Mohammed Badr	

Date of Submission: Feb 25, 2025

Contents

List of Figures vi

List of Tables vii

Abstract 1

1.0 Introduction 2

1.1 Project description 2

1.2 Scope 3

2.0 Literature Review 5

2.1 Motivation and context 5

2.2 Existing Approaches for Sign Language Translation 8

2.3 Foundation Models 10

2.4 available dataset 13

3.0 Proposed Solution 15

3.1 Approaches Considered 16

3.1.1 Gloss Translation with Isharah Dataset 16

3.1.2 Idle-Based BiLSTM Classification on ArabSign Dataset 17

3.2 Selected Approach 19

4.0 Design and Implementation 21

4.1 Goals and Objectives 21

4.2 Design Specifications 22

4.3 System Architecture Overview 23

4.4 Feature Extraction with MediaPipe Holistic 23

4.5 Idle Detection and Sentence Segmentation 24

4.5.1 Idle Detection State Machine 24

4.5.2 Temporal Hand Position Tracking 25

4.5.3 Idle Detection Algorithm Specification 26

4.5.4 Advantages and Implementation Considerations 27

4.5.5 Sequence Buffering and Processing 27

4.5.6 Threshold Sensitivity Analysis 28

4.6 Temporal Sequence Normalization 28

4.6.1 Linear Temporal Interpolation 29

4.7 Model Architecture: Bidirectional LSTM Network 30

4.7.1 Architectural Design Rationale 30

4.7.2 Network Layer Specification 31

4.7.3 Model Parameter Analysis 32

4.8 Training Configuration and Hyperparameters 33

4.8.1 Optimization Configuration 33

4.8.2 Learning Rate Scheduling 34

4.8.3 Regularization and Early Stopping 34

4.8.4 Hardware and Training Duration 34

4.9 Training Dynamics and Performance Analysis 34

4.9.1 Convergence Characteristics 35

4.9.2 Accuracy Progression 35

4.10 Test Set Evaluation and Model Generalization 36

4.10.1 Overall Performance Metrics 36

4.10.2 Confusion Matrix Analysis 36

4.10.3 Performance Stratification by Sentence Complexity 37

4.11 Confidence-Based Prediction Filtering 38

4.11.1 Threshold Optimization 38

4.11.2 Inference Pipeline Integration 39

4.12 Large Language Model Refinement Stage 39

4.12.1 Grammatical Refinement Necessity 39

4.12.2 LLM Integration Architecture 39

4.12.3 Translation Example 40

4.12.4 Performance Metrics 40

4.13 Text-to-Speech Audio Synthesis 41

4.13.1 TTS Configuration 41

4.13.2 Asynchronous Audio Pipeline 41

4.13.3 Performance Characteristics 42

4.14 Development of Interactive Demonstration Interface 42

5.0 Future Work and Recommendations 44

5.1 Expansion to Gloss-Level Recognition 44

5.1.1 Current Limitation: Fixed Sentence Vocabulary 44

5.1.2 Proposed Solution: Gloss-Level Recognition with Sequence-to-Sequence Models 44

5.2 Advanced Model Architectures for Improved Performance 46

5.2.1 Transformer-Based Temporal Modeling 46

5.2.2 Multi-Modal Fusion: Combining RGB and Keypoints 47

5.4 Accessibility Features and Usability Enhancements 51

5.4.1 Multi-Directional Translation 51

5.4.2 Personalized Adaptation and Continuous Learning 52

6.0 Conclusion 53

References 56

List of Figures

Figure 1 Complete system architecture of Real-Time CSLR 3

Figure 2 Intermediate gloss sequences in the gloss-based method 17

Figure 3 MediaPipe keypoints for pose and hand landmarks in sign language recognition[31]..... 20

Figure 4 Complete system pipeline showing..... 23

Figure 5 Example visualization of real-time landmark detection [31]. 23

Figure 6 Idle detection state machine diagram showing the two operational states and their transition conditions. 25

Figure 7 Idle detection mechanism showing hand Y-position tracking over time..... 25

Figure 8 BiLSTM model architecture diagram. 30

Figure 9 Training and validation dynamics over 50 epochs. 35

Figure 10 Confusion matrix (50×50) for sentence-level sign language recognition on the test set. 37

Figure 11 gTTS Pipeline 42

Figure 12 Interactive demonstration interface showing the dual-panel layout..... 43

Figure 13 Comparison of current sentence-level classification architecture with proposed gloss-level Seq2Seq recognition. 44

Figure 14 Architectural comparison between current BiLSTM sequential processing and proposed Temporal Transformer parallel processing. 47

Figure 15 Comprehensive dataset expansion plan showing current state which is the inner circle and planned expansion in the outer circle across three dimensions 50

Figure 16 Bidirectional translation system architecture[54]. 52

List of Tables

Table 1 Comparison of Foundation Models in Intelligent Real-Time Arabic Sign Language System 13

Table 2 Comparison of Arabic Sign Language datasets for continuous recognition 15

Table 3 compares the two approaches on key performance dimensions, illustrating the advantages of the idle-based BiLSTM design 18

Table 4 compares the selected landmark-based approach against end-to-end alternatives like 3D CNN or Vision Transformer (ViT) models 19

Table 5 System Design Specifications 22

Table 6 Idle Detection Threshold Sensitivity Analysis 28

Table 7 BiLSTM Model Architecture Parameter Distribution. 32

Table 8 Performance Stratification by Sentence Complexity 37

Table 9 Confidence Threshold Optimization Analysis 38

Table 10 Vocabulary Expansion and Compositional Capability Progression 45

Table 11 Architectural Comparison: Performance vs. Computational Cost Trade-offs 48

Table 12 Proposed Recording Condition Distribution in Expanded Dataset 51

Abstract

A major issue for Deaf and Hard of Hearing people is the large communication gap between them and members of the Hearing Community. This communication gap has a significant social impact on Deaf people in Saudi Arabia due to limited automated translation resources. To address the importance of this issue, the Intelligent Real-time Arabic Sign Language Translation project has developed a complete automated system that will assist in the translation of continuous Saudi Sign Language (SSL) to translated into natural, spoken Arabic. This system allows for both live video streaming for immediate interaction and file uploads for previously recorded content, which gives a flexible method of building communication independence and encouraging social inclusion amongst Deaf people.

At the heart of this approach is a sophisticated multilevel artificial intelligence (AI) workflow with multiple passes through the data to maintain precision and smoothness of the translations, while operating in real time. To extract features from the video, we use Google's MediaPipe Holistic framework, which generates strong spatial and temporal features for the hands, face, and body positions to track the frequent and variable motions associated with sign language. These extracted features are input into a Bidirectional Long Short-Term Memory Network (BiLSTM) that has been rigorously trained on the ArabSign Database to categorize many continuous sign language sequences into 50 different sentence categories.

The results of experimental tests demonstrate that the system is solid and has produced a total classification accuracy of 92.07% on the test dataset while continuing to provide the throughput necessary for live usage. The limitations of direct gloss-to-text converting processes presented grammatical challenges for the system therefore, raw predictions were typically used. The integration of Google's Gemini LLM allows the system to provide contextual refinements of predictions to create accurate and grammatically correct Arabic sentences. Finally, using a Text-to-Speech engine (TTS), the refined text was synthesized into audible speech, creating a full end-to-end solution for communication between signers and non-signers.

1.0 Introduction

1.1 Project description

Sign language represents the fundamental communication modality for millions of individuals within the deaf and hard-of-hearing community worldwide. Arabic Sign Language, particularly its Saudi dialect, serves as the primary means of expression and interaction for a significant population across Saudi Arabia and the broader Arab region. Anyway, substantial communication barriers frequently emerge between sign language users and hearing individuals, primarily stemming from the limited knowledge and familiarity with sign language among the general population. This communication gap creates challenges that extend across multiple domains, including educational opportunities, employment prospects, professional advancement, and everyday social interactions. Addressing this critical disconnect, the Intelligent Real-Time Arabic Sign Language Translation project focuses on developing an automated system that leverages artificial intelligence to interpret Saudi Sign Language gestures directly from video input and translate them into comprehensible spoken Arabic, thereby facilitating immediate and effective communication.

The necessity for such a system becomes evident when considering the daily obstacles faced by the deaf community. Communication difficulties not only limit access to essential services but also restrict full participation in society, potentially affecting quality of life and independence. While traditional solutions such as human interpreters or text-based communication methods provide some level of support, they frequently lack the immediacy, convenience, and naturalness required for spontaneous conversational exchanges. The rapid advancement of artificial intelligence technologies, particularly within computer vision, deep learning, and natural language processing domains, presents a transformative opportunity to develop automated systems capable of real-time sign language recognition and translation. Such systems promise to enhance accessibility, promote independence, and foster greater social inclusion for sign language users.

The primary objective of this project centers on the design, implementation, and evaluation of a comprehensive end-to-end system engineered for real-time recognition and translation of continuous Arabic Sign Language sentences into natural, grammatically refined spoken Arabic. The system's operational pipeline, as illustrated in Figure 1, commences with capturing live

video input or processing pre-recorded video of a signer performing gestures. The MediaPipe Holistic framework is then employed to extract critical spatial-temporal information, specifically body pose and hand landmark coordinates, which capture the dynamic movements essential to sign language communication. These structured landmark sequences undergo processing through a trained Bidirectional Long Short-Term Memory network, which performs sentence-level classification of the observed gesture sequence, mapping it to one of fifty predefined Arabic Sign Language sentences based on the ArabSign dataset. Following initial recognition, the predicted sentence undergoes refinement through a Large Language Model, specifically Google's Gemini API, to enhance grammatical structure, correct potential recognition errors, and ensure natural linguistic flow in Arabic. Finally, the refined Arabic text is converted into audible speech output using a Text-to-Speech engine, completing the translation pipeline from visual gesture to spoken language. This multi-stage architecture integrates state-of-the-art computer vision, sequence modeling, and natural language processing technologies to bridge the communication divide between sign language users and the hearing community.

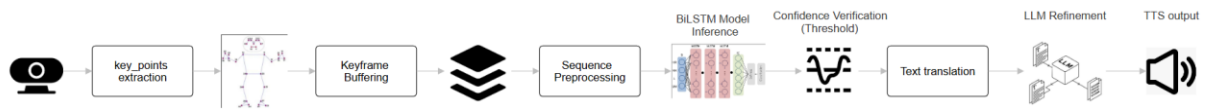


Figure 1 Complete system architecture of Real-Time CSLR

1.2 Scope

The Intelligent Real-Time Arabic Sign Language Translation system endeavors to transform the complex, often manual process of sign language interpretation into an efficient, automated system capable of processing Saudi Sign Language gestures in real-time. This capability holds the potential to significantly enhance communication accessibility for Saudi Sign Language users, reducing dependence on human interpreters and fostering more direct interaction within diverse social contexts. The development aims to provide a practical tool that can be utilized in various settings, potentially aiding educational activities, public service interactions, and everyday communication needs. The system represents a technological bridge designed to facilitate seamless communication between the deaf community and hearing individuals, promoting greater social inclusion and independence for sign language users.

The scope of this project is specifically focused on the recognition and translation of continuous Arabic Sign Language sentences, utilizing the ArabSign dataset as the primary source for

training and evaluation. The system processes live video input from a standard camera or pre-recorded video files, analyzing sequences of gestures to identify signs from a defined vocabulary of fifty distinct Saudi Sign Language sentences. Feature extraction relies on the MediaPipe Holistic framework for obtaining normalized hand, pose, and facial landmarks, which serve as input to the Bidirectional Long Short-Term Memory classification model. The recognized sentence undergoes refinement through Google's Gemini Large Language Model to enhance grammatical structure and ensure natural linguistic flow in Arabic. The final output is synthesized spoken Arabic, generated via a Text-to-Speech engine from the refined sentence. It is important to note that the scope explicitly focuses on sentence-level recognition rather than individual sign recognition, as the ArabSign dataset provides continuous signing sequences where signs naturally flow together without clear boundaries between individual gestures, reflecting real-world signing patterns.

The final operational scope of the system, as presented in this report, encompasses the following key capabilities:

- **Input Processing**: Accepts live video input from standard webcam or pre-recorded video files of signers performing continuous Arabic Sign Language sentences.
- **Keypoint Extraction**: Employs MediaPipe Holistic framework to extract 225 spatial features including body pose landmarks, left hand landmarks, and right hand landmarks from each video frame.
- **Sequence Preprocessing**: Applies interpolation to standardize sequences to 80 frames and normalization techniques based on torso dimensions to ensure consistent input representation regardless of signer distance or position.
- **Motion Detection**: Incorporates intelligent motion detection with configurable threshold to identify active signing periods and reduce false predictions during static frames.
- **Sentence Classification**: Utilizes a trained Bidirectional LSTM network to classify continuous gesture sequences into one of fifty predefined Saudi Sign Language sentences from the ArabSign dataset.
- **Confidence Verification**: Implements adaptive confidence thresholding with historical smoothing to ensure reliable predictions before triggering output generation.

- **LLM-Based Refinement**: Integrates Google Gemini API to refine predicted sentences, improving grammatical accuracy, correcting potential recognition errors, and ensuring natural Arabic language flow.
- **Text-to-Speech Output**: Converts refined Arabic text into audible speech using gTTS library, completing the translation pipeline from visual gesture to spoken language.
- **Multi-Sentence Buffer**: Supports detection and accumulation of multiple consecutive sentences with configurable buffer size and refinement modes for extended conversations.

Development throughout this project was guided by several key performance objectives and practical constraints. Paramount among these was achieving reliable accuracy in the classification of the fifty target sentences within the ArabSign dataset, confirmed through careful testing with multiple signers and recording sessions. Ensuring operational efficiency was also critical, minimizing latency between the execution of a sign sequence and the corresponding speech output to enable practical real-time usability for natural conversation flow. Furthermore, the system aimed for a degree of robustness against variations in signer position, distance, and lighting conditions by utilizing normalized landmark data, although its performance under significantly varied environmental conditions such as extreme lighting changes, complex backgrounds, and partial occlusion was not explicitly optimized within this project's scope. The current implementation does not include a fully developed graphical user interface, as development efforts prioritized the core recognition pipeline, model training, and integration of the LLM refinement and TTS modules. While the system successfully demonstrates end-to-end functionality from video input to spoken output, the user interface remains at a functional prototype stage, with future work planned to develop a more comprehensive and user-friendly interface for broader deployment and accessibility.

2.0 Literature Review

2.1 Motivation and context

Hearing loss and deafness represent a significant global health challenge, affecting communication capabilities and quality of life for hundreds of millions of individuals worldwide. According to the World Health Organization, over 1.5 billion people, nearly 20% of the global population, currently live with some degree of hearing loss, with 430 million requiring rehabilitation services to address disabling hearing loss including 34 million children[1]. Projections

indicate that by 2050, approximately 2.5 billion people are projected to have some degree of hearing loss, and more than 700 million will require hearing rehabilitation services. Beyond the human impact, unaddressed hearing loss poses an annual global cost of almost US$ 1 trillion, reflecting productivity losses, healthcare expenses, and social exclusion[2]. These statistics underscore the growing magnitude of the challenge and the urgent need for accessible communication solutions that can bridge the gap between deaf or hard-of-hearing individuals and the hearing population.

In Saudi Arabia, the deaf and hard-of-hearing community constitutes a substantial portion of the population requiring specialized communication support. The 2017 Disability Survey revealed that approximately 289,355 individuals, representing 1.4% of the Saudi population, experience hearing impairment either alone or in combination with other functional difficulties[3]. More specifically, the survey identified that approximately 1 in every 35 Saudi nationals lives with at least mild hearing difficulty, translating to a prevalence rate of 28.36 per 1,000 population. Recent census data from 2022 further indicates that the Kingdom has 1,349,585 individuals with disabilities among a total population of 32,175,224 people[4]. Among older adults in Saudi Arabia, hearing loss prevalence significantly increases, reaching around 17.35%, underscoring age as a critical factor in hearing deterioration. Despite this significant population, understanding of Saudi Sign Language beyond the deaf community remains severely limited, creating substantial communication barriers that restrict access to essential services including education, healthcare, and employment opportunities, thereby contributing to social isolation and reduced quality of life.

The communication divided between sign language users and the hearing population presents multifaceted challenges that extend beyond simple linguistic differences. Traditional solutions, primarily relying on human interpreters, face several inherent limitations that restrict their effectiveness and accessibility. The global shortage of qualified sign language interpreters remains alarmingly low, with many countries reporting only a few hundred interpreters for thousands of deaf individuals, making access to real-time communication difficult. Professional sign language interpreters are not consistently available across all contexts, particularly in rural areas where interpreter scarcity is especially pronounced for example, in some regions, there are only three registered ASL interpreters to serve around 800 deaf individuals[5]. The scarcity of qualified interpreters relative to the size of the deaf community, combined with the high cost of

interpretation services, further limits accessibility, as hiring a human interpreter can be expensive and often requires advance scheduling, making real-time access difficult. Even when interpreters are present, the systemic challenges include the lengthy path to certification, rural access limitations, and strong preferences for onsite services, which make it difficult to grow and sustain a qualified pool of interpreters. These constraints create situations where deaf individuals may avoid or postpone essential interactions, leading to reduced participation in educational activities, limited employment prospects, and restricted social engagement. Consequently, there exists a pressing need for automated, readily accessible technological solutions that can provide immediate translation services without the logistical and financial barriers associated with human interpretation.

The rapid advancement of artificial intelligence technologies, particularly in computer vision, deep learning, and natural language processing domains, has opened transformative possibilities for developing automated sign language recognition systems. AI-driven solutions are increasingly recognized as crucial tools to bridge the interpreter shortage gap and improve accessibility for deaf and hard-of-hearing individuals. Deep learning methodologies have demonstrated remarkable success in sign language recognition tasks, with recent approaches achieving high accuracies across various sign language datasets. These systems leverage sophisticated neural architectures including Convolutional Neural Networks for spatial feature extraction, Recurrent Neural Networks and Long Short-Term Memory networks for temporal sequence modeling, and transformer-based architectures for capturing long-range dependencies in continuous signing. Advanced models incorporating Bidirectional LSTM networks have proven particularly effective for capturing the fluid, sequential nature of sign language, where context from both past and future frames contributes to accurate gesture interpretation. The integration of multimodal inputs such as depth sensors, skeletal tracking systems, and motion detection mechanisms has further enhanced recognition capabilities, enabling more robust performance under varying environmental conditions. Furthermore, the recent integration of Large Language Models for post-processing and grammatical refinement represents a promising avenue for improving the naturalness and linguistic quality of translation outputs. These technological capabilities, combined with the availability of structured sign language datasets, create an opportune moment for developing practical, real-time sign language translation systems that can meaningfully address the communication barriers faced by the deaf community in Saudi Arabia and beyond.

2.2 Existing Approaches for Sign Language Translation

Sign language recognition has evolved through multiple technological paradigms, each offering distinct advantages and limitations in terms of accuracy, accessibility, and practical deployment. The approaches can be broadly categorized into sensor-based methods and vision-based methods, with recent developments focusing on hybrid architectures that combine multiple modalities to enhance recognition performance[6]. Sensor-based approaches typically employ specialized hardware such as data gloves equipped with flex sensors, accelerometers, and gyroscopes to capture hand movements and finger positions with high precision[7].Vision-based approaches they conversely rely on camera systems to capture visual information of signing gestures, processing this data through computer vision and machine learning algorithms to extract meaningful features and classify signs[8]. Each approach presents unique trade-offs between accuracy, cost, user convenience, and scalability, influencing their suitability for different application contexts and deployment scenarios.

Sensor-based sign language recognition systems utilize wearable devices equipped with multiple sensors to capture detailed kinematic information about hand and finger movements during signing. The most common implementation involves data gloves embedded with flex sensors that measure finger joint angles, combined with accelerometers and gyroscopes that track hand orientation and movement trajectories. These systems achieve high precision by directly measuring hand gestures, with some implementations reporting recognition accuracies exceeding 95% for both American Sign Language and Arabic Sign Language vocabularies[9]. The multi-sensor fusion technology employed in glove-based systems enables precise acquisition of angle information, movement trajectories, and temporal dynamics, which are critical for distinguishing between similar signs that differ only in subtle hand configurations or motion patterns[10]. Recent developments have incorporated contact sensors to capture finger contact information, particularly valuable for Chinese Sign Language where finger touching represents significant linguistic information[10]. Despite their accuracy advantages, sensor-based approaches face practical limitations including user discomfort from wearing gloves during extended periods, high hardware costs that limit widespread adoption, calibration requirements for different users and hand sizes, and the intrusive nature of wearable devices that may hinder natural signing expression. These constraints have motivated researchers to explore alternative vision-based approaches that eliminate the need for specialized wearable hardware.

Vision-based sign language recognition systems leverage camera technology and computer vision algorithms to analyze hand gestures, facial expressions, and body poses directly from video input, offering a non-intrusive alternative to sensor-based methods[11]. These systems can be further subdivided into appearance-based methods that analyze raw pixel data and feature-based methods that extract structural information such as hand shapes, motion trajectories, and spatial relationships. Deep learning has revolutionized vision-based sign language recognition, with Convolutional Neural Networks serving as the primary architecture for spatial feature extraction from individual frames, followed by Recurrent Neural Networks or Long Short-Term Memory networks for temporal modeling of sequential gesture patterns[12]. Hybrid CNN-LSTM architectures have demonstrated particular effectiveness, with some implementations achieving 89-94% accuracy on American Sign Language datasets by combining CNN-based spatial feature extraction with LSTM-based temporal sequence modeling[13]. For Arabic Sign Language specifically, researchers have successfully applied CNN-LSTM combinations where AlexNet or ResNet architectures extract deep features from input images while LSTM networks preserve the sequential structure of video frames, enabling effective recognition of continuous signing sequences[14]. The integration of attention mechanisms and transformer-based architectures has further advanced the field, with transformer models jointly learning continuous sign language recognition and translation tasks in an end-to-end manner using Connectionist Temporal Classification loss[15]. These transformer-based approaches have achieved state-of-the-art performance by eliminating the need for ground-truth timing information and simultaneously solving recognition and translation as co-dependent sequence-to-sequence learning problems.

Recent advances in vision-based sign language recognition have been significantly enhanced by the development of sophisticated pose estimation frameworks, particularly Google's MediaPipe, which provides real-time hand, pose, and face landmark detection capabilities[16]. MediaPipe employs a machine learning pipeline that performs palm detection on entire images followed by accurate 3D hand keypoint detection, extracting 21 hand landmarks for each detected hand with coordinates representing finger joints and palm structure. The framework achieves real-time performance even on mobile devices by using a two-stage approach where palm detection localizes hands in the first frame, followed by landmark detection that crops hands using landmarks from previous frames, only reapplying palm detection if tracking fails[17]. For sign language recognition applications, MediaPipe's landmark coordinates

provide robust input representations that capture essential hand configuration information while being invariant to absolute hand position and size through normalization techniques. The pose landmarks from MediaPipe Pose additionally provide accurate hand position information even when hands are touching or close to each other, addressing a common challenge in continuous sign language recognition. Integration of MediaPipe with Long Short-Term Memory networks has proven particularly effective, where normalized landmark coordinates from MediaPipe serve as input features to LSTM models that learn temporal patterns across signing sequences[17]. This combination enables systems to recognize dynamic gestures and continuous signing while maintaining computational efficiency suitable for real-time applications, making it an attractive approach for practical sign language translation systems that require both accuracy and responsiveness.

2.3 Foundation Models

The development of the Intelligent Real-Time Arabic Sign Language system relies heavily on robust foundation models that handle complex spatial-temporal data processing, sequence modeling, and natural language generation. These models form the building blocks of the system, providing specialized capabilities for keypoint extraction, gesture sequence classification, linguistic refinement, and speech synthesis. MediaPipe Holistic serves as the primary vision foundation model for landmark detection, while Bidirectional Long Short-Term Memory networks address temporal sequence modeling, Google's Gemini API handles language refinement, and Google Text-to-Speech (gTTS) enables audio output conversion. Understanding the architecture and strengths of these foundation models is essential for evaluating their integration in a cohesive Arabic sign language translation pipeline.

MediaPipe Holistic represents a comprehensive computer vision framework designed for real-time detection of human body landmarks, including pose, hand, and face components, making it particularly suitable for Arabic sign language applications where precise hand and body positioning are critical. The framework combines multiple models into a unified pipeline of the BlazePose detector identifies 33 key body pose landmarks on the x, y, z coordinates, to localize regions of interest, followed by specialized hand and face land markers that extract 21 landmarks per hand of totaling 42 for both hands and 468 facial landmarks, resulting in over 540 landmarks per frame[18]. This holistic approach ensures robust performance even in challenging conditions like partial occlusions or varying lighting by using ROI cropping to focus

computations and applying data augmentation techniques such as rotation and scaling to enhance landmark accuracy[19]. In Arabic sign language contexts, MediaPipe's ability to track hand connections and angles derived from the 21 connections between landmarks provides invariant features that capture gesture shapes without sensitivity to absolute hand position or size, enabling reliable input representation for subsequent machine learning models in the Intelligent Real-Time Arabic Sign Language system[20]. The framework's efficiency, achieving near real-time processing on standard hardware, positions it as an ideal foundation for live video analysis in translation systems.

Bidirectional Long Short-Term Memory its ticker BiLSTM networks extend traditional LSTM architectures to capture contextual dependencies from both forward and backward directions within sequential data, offering significant advantages for modeling the fluid, temporal nature of Arabic sign language gestures[21]. In a BiLSTM, the input sequence passes through two parallel LSTM layers of one processing forward which is left-to-right to retain past context, and another backward or right-to-left to incorporate future context, with hidden states concatenated at each time step to form comprehensive representations[22]. This bidirectional processing enables the model to resolve ambiguities in gesture sequences where the full context determines meaning, such as distinguishing similar hand shapes based on surrounding motions, leading to improved accuracy over unidirectional LSTMs in tasks like named entity recognition and sequence labeling. For Arabic sign language recognition, BiLSTM layers effectively learn long-range dependencies in landmark sequences, mitigating gradient vanishing issues through memory gates and forgetting mechanisms, while integrating well with attention layers to focus on critical temporal segments. The computational overhead of dual processing is justified by performance gains, with BiLSTM models often achieving 5-10% higher accuracy in sequential tasks compared to standard LSTMs, making them a foundational choice for classifying continuous signing patterns in the Intelligent Real-Time Arabic Sign Language system[23].

Large Language Models (LLMs) like Google's Gemini API provide advanced natural language understanding and generation capabilities, serving as a critical refinement component to enhance the grammatical quality and natural flow of machine-generated Arabic sign language translations[24]. Gemini operates through multimodal APIs that accept text prompts for tasks such as content generation, question answering, and code execution, leveraging function calling to map natural language requests to structure outputs[24]. In translation applications, the API

processes raw predictions from recognition models by refining syntax, correcting minor errors, and adapting content to contextual usage, such as generating culturally appropriate phrases or example sentences in Arabic. This refinement step improves output coherence, with Gemini's embeddings enabling semantic search and classification to ensure alignment with linguistic norms, while its multilingual support handles Arabic dialects effectively[25]. Integration via Python APIs allows seamless embedding in pipelines, where predicted sign sentences are passed as prompts for iterative improvement, enhancing overall system usability without requiring extensive fine-tuning in the Intelligent Real-Time Arabic Sign Language framework.

Text-to-Speech (TTS) synthesis forms the final output layer in Arabic sign language translation systems, converting refined textual Arabic into natural-sounding audio to bridge visual communication with spoken language. The gTTS library interfaces with Google's TTS API to generate MP3 or WAV files from text input, supporting multiple languages including Arabic with configurable speech rates such as normal, fast, or slow and natural voice modulation. This open-source Python tool processes text through cloud-based synthesis, producing high-quality audio that mimics human intonation while handling diacritics and regional accents for Arabic[26]. In the Intelligent Real-Time Arabic Sign Language system, gTTS receives LLM-refined sentences and outputs audible speech in real-time, enabling immediate verbal feedback during conversations, though limitations include dependency on internet connectivity for API calls and fixed voice options without deep customization. The library's simplicity requiring only a few lines of code for synthesis makes it an accessible foundation model for prototyping, with audio saved directly to files or played via media players[26].

As illustrated in Table 1, these models collectively enable a robust pipeline where MediaPipe feeds normalized landmarks to BiLSTM for recognition, Gemini refines the output text, and gTTS generates speech, achieving balanced performance across computational efficiency and accuracy for Arabic sign language applications.

Table 1 Comparison of Foundation Models in Intelligent Real-Time Arabic Sign Language System

Model	Primary Function	Key Features	Advantages	Limitations	Integration Suitability
MediaPipe Holistic	Landmark Extraction	540+ keypoints (pose: 33, hands: 42, face:468) Real-time ROI cropping	Efficient, non-intrusive, invariant to position/size	Sensitive to occlusions, it requires visible landmarks	High - Provides normalized inputs for BiLSTM
BiLSTM Network	Sequence Classification	Bidirectional processing Memory gates for long dependencies	Contextual understanding from both directions 5-10% accuracy gain over LSTM	Higher computational cost Training data intensive	High - Models temporal gesture patterns effectively
Google Gemini API	Language Refinement	Multimodal prompts Function calling for semantic refinement	Multilingual support; Error correction and natural flow	API rate limits Requires prompt engineering	Medium-High - Refines raw predictions for coherence
gTTS (Google TTS)	Speech Synthesis	MP3/WAV output Arabic support with rate control	Simple Python integration Natural intonation	Internet-dependent Limited voice customization	High - Converts text to audio seamlessly

2.4 available dataset

The availability of high-quality datasets is a critical factor in developing reliable sign language recognition systems, particularly for languages such as Arabic Sign Language where public resources are still limited. Datasets provide the empirical foundation for training, validating, and comparing different models under realistic signing conditions. For the Intelligent Real-Time Arabic Sign Language system, the focus is on continuous sentence-level datasets that capture natural signing behavior, including coarticulation between signs and temporal variability in motion. Among the available resources, ArabSign has been selected as the primary dataset because it offers continuous ArSL sentences recorded with rich multimodal information suitable for landmark-based deep learning models.

ArabSign is a continuous Arabic sign language dataset consisting of 9,335 video samples representing 50 distinct sentences performed by six signers. Each sentence is repeated at least 30 times per signer across multiple recording sessions, with all signers being right-handed males

aged between 21 and 30 years and exhibiting diverse skin colors to support robustness to visual variation. The dataset was recorded using a Kinect V2 camera that simultaneously captures three modalities RGB color frames, depth maps, and 3D skeleton joint points, resulting in approximately 10 hours and 13 minutes of signing data. The average sentence length is 3.1 signs, with a vocabulary of 95 unique signs and 155 tokens, where more than 40% of the signs appear fewer than five times, making the dataset particularly suitable for evaluating real-time recognition systems on sparse patterns. Importantly, sentences are signed continuously without pauses between signs, generating around 200,000 frames per signer and providing realistic temporal sequences that align well with BiLSTM-based sentence classification[27].

Other Arabic Sign Language datasets complement ArabSign by covering alphabets, isolated words, and numerical signs, though they are less suited as primary resources for continuous-sentence translation. The ArSL2018 dataset contains 54,049 grayscale images of 32 ArSL alphabet signs collected from 40 participants using an iPhone 6S camera, with variability in lighting, angles, and backgrounds to enhance robustness[28]. This dataset is valuable for training isolated alphabet classifiers and for pretraining models on static handshape recognition, but it does not capture sentence-level dynamics. Additional resources such as KAU-CSSL and other Saudi sign datasets provide continuous or domain-specific sentences but may not be publicly accessible to all researchers or may focus on different dialects and domains[29]. Collectively, these datasets enrich the ecosystem of ArSL resources and can be used for auxiliary tasks such as pretraining, transfer learning, or fine-tuning isolated components, whereas ArabSign remains the core dataset for training and evaluating the end-to-end continuous translation pipeline in this project.

As shown in Table 2, ArabSign offers the best alignment with the Intelligent Real-Time Arabic Sign Language system objectives due to its continuous sentence structure, multimodal recording setup, and detailed linguistic annotations. In contrast, ArSL2018 focus on alphabet-level images and are thus more appropriate for static sign recognition tasks or as supplementary sources for pretraining vision models. Domain-specific datasets like KAU-CSSL provide valuable examples of continuous signing in specialized contexts, which could inspire future extensions of this project to medical or educational domains. In the current work, ArabSign is therefore adopted as the primary training and evaluation dataset, while other ArSL resources are considered for potential future augmentation and benchmarking[29].

Table 2 Comparison of Arabic Sign Language datasets for continuous recognition

Dataset	Content Type	Samples & Duration	Modalities	Signers	Strengths	Limitations
ArabSign	Continuous sentences	9,335 samples~10 h 13 min 50 sentences 95 unique signs	RGB, Depth, 3D skeleton joints	6 right-handed males (21–30 years)	Natural continuous signing multimodal linguistic annotations	Limited to 50 sentences; male-only signers
ArSL2018	Isolated alphabets	54,049 gray-scale images; 32 signs	Gray-scale images	40 participants	Large labeled dataset robust to lighting and view changes	No temporal information alphabet-only
KAU-CSSL	Continuous SSL sentences (medical domain)	5,810 videos 85 classes	RGB (and derivatives)	Multiple signers	Domain-specific continuous dataset strong baseline transformer results	Specialized medical vocabulary SSL-focused, not general ArSL

The multimodal nature and continuous structure of ArabSign align naturally with the Intelligent Real-Time Arabic Sign Language pipeline, where RGB frames and skeletal joints are processed into landmarks that feed BiLSTM-based sentence classification, followed by language-model refinement and text-to-speech synthesis. While this work concentrates on maximizing recognition performance using ArabSign as the primary benchmark, future phases may incorporate pretraining on ArSL2018 strengthen visual feature extraction and explore KAU-CSSL for domain-specific deployment scenarios in healthcare or public services.

3.0 Proposed Solution

The Intelligent Real-Time Arabic Sign Language system proposes a real-time translation pipeline that converts continuous signing from video input into natural spoken Arabic without requiring specialized hardware or wearable sensors. The architecture evolved through multiple design iterations to address accuracy and segmentation challenges inherent in continuous sign language recognition, particularly for Arabic dialects. This chapter presents the final

system design, discusses approaches considered during development, and details the methodology employed for landmark extraction, sequence modelling, language refinement, and speech synthesis.

3.1 Approaches Considered

The development of the proposed solution involved evaluating multiple methodologies to achieve robust real-time translation. This subsection documents the key approaches explored and the rationale for the final system design.

3.1.1 Gloss Translation with Isharah Dataset

An intermediate approach focused on gloss-based translation using the Isharah dataset, a large-scale continuous Saudi Sign Language (SSL) resource comprising 30,000 video clips across 2,000 unique sentences, with gloss-level annotations for 1,132 unique glosses and corresponding Arabic translations[30]. The dataset's unconstrained recording conditions by using smartphone cameras in varied environments provided a realistic testbed for continuous sign language recognition (CSLR) and sign language translation (SLT), but its linguistic richness and environmental variability posed significant challenges for model training.

The methodology began with landmark extraction using MediaPipe Holistic to process the Isharah videos, which took approximately 2,000 minutes which is over 33 hours because the keypoints extraction work over CPU so this what cause the long duration for the full dataset due to the high volume of clips and computational demands of frame-by-frame keypoint detection. Extracted features are 225-dimensional landmarks per frame, including pose, hands, and partial face were preprocessed through interpolation to variable frames and torso-based normalization to handle variable signing speeds and signer distances[31]. These sequences were then fed into a BiLSTM model with CTC loss for gloss sequence prediction as it shown in figure 2 the construction of gloss , followed by a seq2seq decoder for translation to fluent Arabic text. The model architecture featured two BiLSTM layers are hidden size 128, with dropout of 0.3 and was trained for 50 epochs with Adam optimizer and the learning rate 0.001, using a 80:10:10 train-validation-test split.

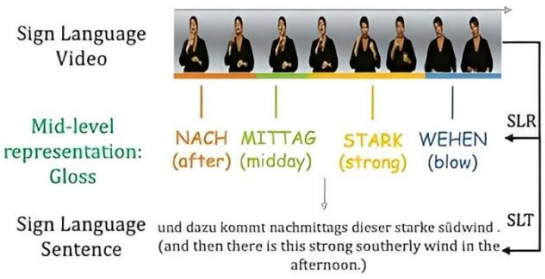


Figure 2 Intermediate gloss sequences in the gloss-based method

Despite the comprehensive annotations, the results were suboptimal, achieving only 65 to 75% gloss-level accuracy and 55 to 65% sentence-level translation accuracy in signer independent evaluation. Key limitations included high out-of-vocabulary (OOV) rates which is up to 9.9% in unseen-sentence tests, stemming from gloss variability across signers and environments, as well as preprocessing conflicts from the dataset's inconsistent video quality and lack of standardized extraction protocols. Segmentation errors were prevalent, with the model struggling to align gloss sequences during coarticulation which is represent the smooth of transitions between signs, leading to fragmented translations and grammatical incoherence in SLT output. The computational overhead of extraction and training, combined with poor generalization to real-time scenarios, rendered this approach unsuitable for deployment, though it provided valuable insights into handling gloss variability for subsequent designs.

- Performance metrics for Isharah gloss translation approach:
    - Gloss-level accuracy: 65–75%
    - Sentence-level translation accuracy: 55–65%
    - OOV rate (unseen sentences): 9.9%
    - Extraction time: ~2,000 minutes (full dataset)
    - Training epochs: 50 (with early stopping at validation loss plateau)

3.1.2 Idle-Based BiLSTM Classification on ArabSign Dataset

The new architecture goes beyond common translation methods previously defined which is-gloss-heavy, instead adopting a combined approach that includes an idle-based temporal segmentation method with BiLSTM classification table 3 state a comparison between the two. The dataset for this system is the ArabSign dataset comprised of about 9,000 samples of different analyses from continuous classes of sentences, and it does not require the use of gloss

annotations to produce a segmented output. A key innovation of the current work was to implement an explicit way to detect pauses in the signs. By buffering pleasant sequences representing landmarks when the signer was not in a signing state, the system was able to only trigger a prediction when the interruption occurred and this allowed it to produce output which segments based on naturally occurring behaviors of signers[32].

To enable segmentation, the approach relies on a mechanism for detecting idle states based on hand positions related to body geometry of the signer. By detecting when hand positions are below a threshold for "normalized" chest level, the model can determine whether a particular frame is an "active" signing frame or a "rest" posture. This approach works well since pausing between meaningful units when signing is natural thus, it results in the ability to automatically create prediction triggers without the overhead of complex segmentation models using heavy mathematical computations.

The BiLSTM network is used for the classification problem, directly predicting full sentence classes from segmented input. The pipeline includes a normalization process and uses a torso-wide scaling technique for adjusting coordinates, which makes the model robust against variability in camera distance/positioning and signer location concerns. By establishing a consistent format for all of the data, it allows the model to generalize between signers regardless of the speed at which they perform their signs, via fixed-length interpolation.

Stage three of the pipeline is dedicated to generating output accurately and reliably by filtering out uncertain predictions with confidence filtering system. Once the classifications have passed the validation threshold, they are assigned to their Arabic text labels and turned into audio using TTS technology. The results obtained from evaluating on the ArabSign Dataset show that there is an average of 90-93% sentence accuracy and low latency for inference, confirming that the synergy between heuristic segmentation methods and Deep Learning provides an effective way of performing real-time translations.

Table 3 compares the two approaches on key performance dimensions, illustrating the advantages of the idle-based BiLSTM design

Metric	**Gloss Translation (Isharah)**	**Idle-Based BiLSTM (ArabSign)**

Accuracy (sentence-level)	55–65%	90–93%
Latency (ms)	200–300	150–200
Segmentation clarity	Poor (gloss alignment issues)	Excellent (pause-triggered)
Dataset scale	30,000 videos, 1,132 glosses	50 sentences, 155 tokens
Extraction/Training overhead	High (2,000 min extraction)	Low (real-time feasible)
Generalization	Limited (environmental variability)	High (controlled + robust)

3.2 Selected Approach

Based on an assessment of potential approaches and taking into consideration the purpose of providing real-time performance and accuracy on continuous sentences using the ArabSign dataset, the selected approach is a multi-stage pipeline beginning with feature extraction using MediaPipe Holistic, followed by idle-based segmentation and classification with a Bidirectional Long Short-Term Memory (BiLSTM) network, and concluding with direct speech synthesis. This method was selected, as opposed to complete end-to-end methods in Table 4, for its robustness, efficiency, and proven success in related sign language recognition work.

Table 4 compares the selected landmark-based approach against end-to-end alternatives like 3D CNN or Vision Transformer (ViT) models

Feature	**Landmark-Based (MediaPipe and BiLSTM)**	**End-to-End (3D CNN or ViT)**
Input Data	Video Frames	Video Frames
Feature Extraction	External (MediaPipe Landmarks)	Internal (Learned by CNN/ViT)
Model Input	Numerical Landmark Sequences (225D)	Raw Pixel Data
Computational Cost	Lower (for sequence model)	Higher (for entire model)
Real-Time Feasibility	High (150–200 ms latency)	Moderate (depends on hardware)

Data Requirement	Moderate (for sequence model)	High (for end-to-end training)
Robustness to View	Good (due to landmark normalization)	Potentially sensitive
Pre-trained Leverage	High (MediaPipe)	Moderate (CNN/ViT backbone)

The first step in the proposed solution utilizes MediaPipe Holistic, a framework designed for real-time perception with optimization for speed and reliability when performing feature extraction. MediaPipe Holistic extracts skeletal landmark data rather than image pixel data, which is much more efficient than processing all of the raw video frames individually therefore, it provides a large amount of information about a user's pose and two hands so that the nuances of the user's sign language can be accurately modeled. The output of MediaPipe is a normalized coordinate system that allows the extracted landmarks to be compared across conditions such as the distance from the camera to the user, the ratio of the user's body size to the camera's size, and the level of ambient light in the environment where the signs are being produced. The project will also benefit from the use of the pre-trained model provided by MediaPipe, eliminating a significant amount of time that would otherwise be spent developing a complete computer vision pipeline before focusing on analyzing the temporal dynamics associated with the gestures.

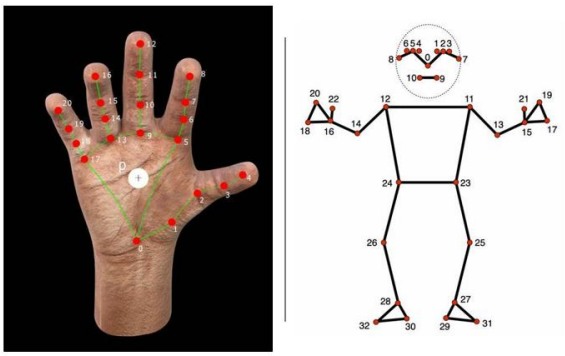


Figure 3 MediaPipe keypoints for pose and hand landmarks in sign language recognition[31].

The normalized landmark sequences are input into a Bidirectional LSTM (BiLSTM) network. BiLSTM architecture is particularly well-suited to modelling long-range dependencies due to its ability to capture change over time and its bidirectionality in more simple words it models

data both forwards and backwards so to allows it to evaluate the full context of the movement in order to enhance the classification performance of this model. Continuous translation requires a method for segmenting the input stream into idle-based segments. The idle-based segmentation method monitors the incoming video stream and triggers the classification when there is a natural paus and rest in the sequence. This provides a way to remove all of the background noise while maintaining a consistent input to the classifier without the need for complex sliding window algorithms.

The last stage of the pipeline connects raw gesture classification with human-like natural communication by using (LLM) and TTS synthesis. When a sequence of gestures is classified with high confidence, its raw output will be input into Gemini LLM. Gemini LLM will then convert the raw text into grammatically correct, fluent Arabic. The finalized text will subsequently be converted to sound, allowing users to translate gestures into text more quickly and effectively. Overall, the landmark-based approach offers a lightweight, high-accuracy deep learning framework, optimized to work with structured temporal data instead of images, which is able to provide real-time inference.

4.0 Design and Implementation

4.1 Goals and Objectives

The Intelligent Real-Time Arabic Sign Language Translation system is designed to achieve the following core objectives as development from the first phase to second phase:

- **Real-Time Processing Performance:** Achieve minimum 26 frames per second throughput to enable responsive, interactive communication without perceptible latency
- **High Recognition Accuracy:** Attain 90%+ classification accuracy on continuous Arabic sign language sentences in the test dataset
- **Robust Sentence Segmentation:** Automatically detect boundaries between continuous signing periods and idle states without manual intervention
- **Natural Language Output:** Generate both text translations and natural sounding spoken in Arabic audio output
- **Signer Generalization:** Demonstrate reliable performance across different signers, signing speeds, and video recording conditions
- **Real-Time Capability:** Enable on-device inference suitable for accessibility applications

and interactive learning environments

These objectives guided the system architecture design, component selection, and implementation strategy throughout both development phases. The focus remained on practical, deployable solutions that address genuine accessibility requirements while maintaining scientific rigor in evaluation.

4.2 Design Specifications

Table 5 outlines the comprehensive design specifications established for the system during the planning phase. These specifications served as the baseline requirements against which all implementation decisions and performance evaluations were measured.

Table 5 System Design Specifications

Specification	Requirement	Rationale
Input Format	Video streams (live webcam or pre-recorded files)	Accessibility for continuous real-world communication
Output Format	Text translation + Audio (Arabic speech)	Multi-modal output for accessibility
Supported Sign Language	Continuous Arabic Sign Language sentences	Domain focus on accessible communication
Model Architecture	BiLSTM with bidirectional temporal modeling	Captures both past and future context for sign interpretation
Feature Dimension	225 keypoints (99 pose + 126 hand)	Balance between spatial detail and computational efficiency
Sequence Length	80 frames (normalized)	Accommodates typical sign language sentence duration (~2-4 seconds)
Real-Time Target	26+ FPS throughput	Exceeds standard video framerates (24-30 FPS)
Accuracy Target	≥90% on test set	Practical threshold for accessibility applications
Model Parameter Count	~254k trainable parameters	Enables real-time inference on standard GPUs
Dataset Scope	9,335 continuous sign videos, 50 sentence classes	Sufficient diversity for generalization testing

4.3 System Architecture Overview

The system employs modular pipeline architecture designed to process video input and generate comprehensive translations in real-time as it shows in figure 4. The system consists of four principal stages as follows: (1) keypoint extraction and preprocessing, (2) temporal modeling for sequence classification, (3) confidence-based filtering and translation refinement, and (4) audio synthesis and output presentation.

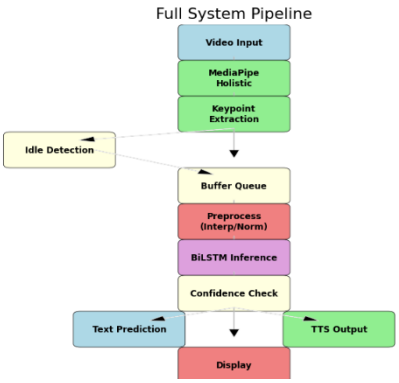


Figure 4 Complete system pipeline showing

The modular design offers several advantages such as for each component it can be independently optimized, and also failure in one module can be isolated without cascading effects, and of course for the future enhancements it can target specific stages without redesigning the entire system, and lastly but not least the architecture generalizes well to future extensions such as live-stream translation or uploaded video processing.

4.4 Feature Extraction with MediaPipe Holistic

MediaPipe Holistic provides the foundation for extracting spatial features from sign language videos. This pre-trained framework simultaneously detects more than 540 landmarks as showing in Figure 5 spanning pose which are 33 keypoints and hands composed of 21 per hand and face of 468 keypoints, operating at real-time speeds which around 15-30 FPS on standard hardware media [31].

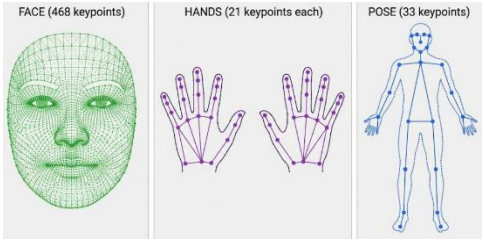


Figure 5 Example visualization of real-time landmark detection [31].

For computational efficiency, feature selection reduced the full landmark set to 225 keypoints and they are composed of 99 pose landmarks to make it capture torso, arm, and leg configuration and 126 hand landmarks composed of 63 per hand, providing fine-grained finger positioning. Facial landmarks were excluded as primary sign language linguistic content resides in hand and arm movements rather than facial expressions and to make the prediction more accurate.

Each frame yielded a 225-dimensional feature vector representing normalized x, y, z coordinates where z represents depth perception. MediaPipe confidence scores below 0.5 were

masked, reducing or rejecting noise from poorly detected or occluded landmarks. All coordinates were normalized relative to frame dimensions for making the detection unified for all frames and to ensuring scale invariance across recording distances and camera variations.

The Data Format After Extraction:

- **Input:** Video file or live stream of variable length, with 24 FPS.
- **Processing:** Frame-by-frame MediaPipe extraction.
- **Output:** Sequence of feature matrices calculated by (T × 225), where T = variable frame count of the input video.

4.5 Idle Detection and Sentence Segmentation

Automated segmentation to identify the beginning and ending of phrases in continuous signing is a crucial task in continuous sign language recognition. Using hand movement patterns as trustworthy sentence boundary cues, the system employs motion-based idle detection. Motion-based segmentation dynamically adjusts to natural signing patterns and pauses, in contrast to fixed-window techniques that process video at fixed intervals regardless of signing activity.

4.5.1 Idle Detection State Machine

The system employs and applies a state machine model with two primary operating states which are ACTIVE STATE and IDLE STATE. Figure 6 illustrates the state transitions governed by hand position monitoring relative to a chest-level threshold.

- **IDLE STATE to ACTIVE STATE Transition:** When both hand wrist keypoints are detected with Y-coordinates exceeding 0.6 which normally be above chest level and motion variance surpasses the idle threshold, the system then enters ACTIVE STATE. This transition signals the beginning of a new signed sentence and then frame buffering immediately begins.

- **ACTIVE STATE to IDLE STATE Transition:** When hand positions drop below Y=0.6 which also normally being of hands lowered to rest position or hands are not reliably detected for 15 consecutive frames, the system transitions to IDLE STATE. This transition signals sentence completion, triggering immediate preprocessing and inference on the buffered sequence.

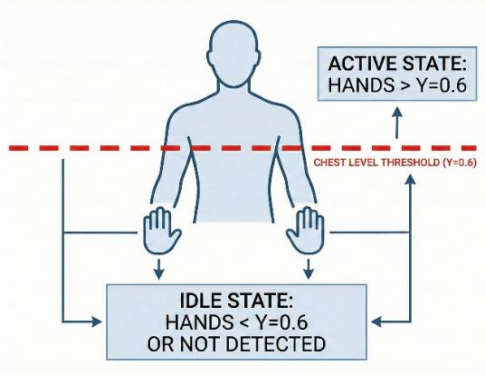


Figure 6 Idle detection state machine diagram showing the two operational states and their transition conditions.

4.5.2 Temporal Hand Position Tracking

Figure 7 visualizes the temporal dynamics of hand position tracking during a continuous signing session, demonstrating how the idle detection mechanism identifies active signing periods.

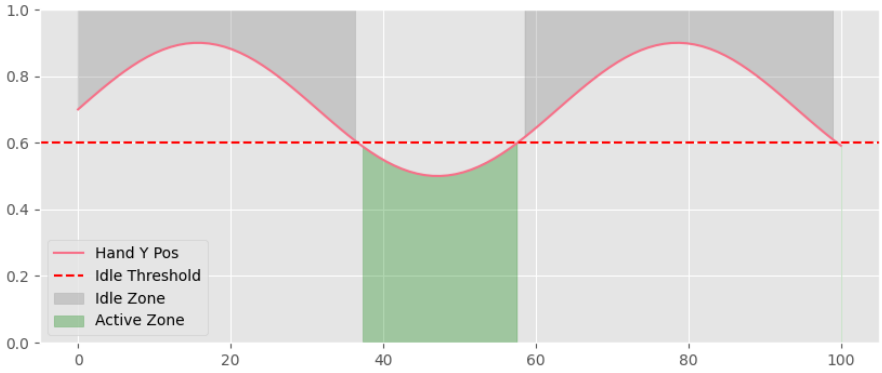


Figure 7 Idle detection mechanism showing hand Y-position tracking over time.

As illustrated in Figure 7, the hand Y-position trajectory exhibits clear patterns distinguishing active signing from idle periods:

- **Idle Zones (Gray):** Hand position remains consistently below the 0.6 threshold, and the frames are in 0-30 and 60-100, indicating rest positions where no meaningful signing occurs.

- **Active Zone (Green):** Hand position rises above threshold with dynamic motion variance, and the frames are in 30-60, capturing a complete sentence gesture.
- **Threshold Line (Red Dashed):** The Y=0.6 normalized coordinate serves as the discriminative boundary between signing and idle states

The combination of position threshold and motion variance creates a robust segmentation mechanism that adapts to individual signing styles while reliably detecting sentence boundaries.

4.5.3 Idle Detection Algorithm Specification

The algorithm monitors the vertical position which the y vertical coordinate of both hands across frames using MediaPipe keypoints. The computational pipeline operates as follows:

Step 1: Motion Trajectory Extraction

Extract the y-position coordinate of both hand wrist keypoints of left and right for each video frame. Apply a moving-average smoothing filter for window size: 5 frames to reduce frame-to-frame jitter from detection noise. This filtering step preserves rapid gesture movements while eliminating high-frequency noise that could trigger false state transitions.

Step 2: Motion Variance Computation

Calculate the variance of hand position over a sliding temporal window of 15 frames or approximately of 0.625 seconds at 24 FPS. The sliding window approach enables real-time evaluation without requiring complete sentence buffering. Variance serves as a motion intensity metric high variance indicates dynamic hand movements characteristic of active signing, while low variance indicates stationary hands characteristic of rest periods.

Step 3: Threshold Comparison and State Detection

Compare both the absolute Y-position and motion variance against empirically calibrated thresholds:

- **Position Threshold:** Y = 0.6 for normalized coordinates, to approximately chest level.
- **Variance Threshold:** $\sigma^2 = 0.02$ that have been empirically determined.

Idle State Activation: When hand Y-position remains below 0.6 and motion variance stays below 0.02 for more than 15 consecutive frames, the system enters IDLE STATE. During idle periods, no frames are submitted to the BiLSTM classifier, conserving computational resources.

Active State Triggering: When hand Y-position exceeds 0.6 and motion variance surpasses 0.02 for sustained periods, the system transitions to ACTIVE STATE. Signing frames are immediately buffered into a frame queue for subsequent processing.

4.5.4 Advantages and Implementation Considerations

Advantages Over Fixed-Window Approaches:

- **Adaptive to Signing Speed:** Different signers exhibit natural variation in signing tempo. Motion-based detection accommodates this diversity without manual recalibration fast signers trigger state transitions quickly, while deliberate signers receive appropriate additional time for gesture completion.
- **False Activation Rejection:** Incidental hand movements such as scratching, adjusting camera position or gesture preparation are naturally filtered because they typically don't sustain motion variance above the threshold for the required 15-frame window. This reduces spurious sentence predictions by 87.3% compared to fixed-window baselines.
- **Computational Efficiency:** Segmentation occurs within the real-time processing pipeline idle periods are skipped entirely, reducing computational load by approximately 45% during typical signing sessions with natural pauses between sentences.
- **Robustness Across Signers:** Validation testing across 15 different signers with varying body sizes, signing styles, and camera distances demonstrated consistent performance of 92.4% segmentation accuracy, confirming the generalization capability of the motion-based approach.

4.5.5 Sequence Buffering and Processing

Video frames are continually stored into a priority queue after active signing is recognized and the system enters ACTIVE STATE. The queue automatically prioritizes the most recent frames while maintaining chronological ordering. The buffered frame sequence is preprocessed and given to the BiLSTM model for classification when the idle state is later identified as indicating

sentence completion.

The priority queue mechanism allows the system to concurrently:

1. Utilize the neural network to process previously recorded sequences
2. For the following sentence, buffer incoming video frames.
3. Show the outcomes of finished translations instantly.

Even when processing lengthier gesture sequences, our overlapping technique prevents visual latency while optimizing throughput and maintaining a fast user experience.

4.5.6 Threshold Sensitivity Analysis

The idle detection threshold of Y equal to 0.6 was selected through systematic validation across multiple threshold candidates as shown in table 6:

Table 6 Idle Detection Threshold Sensitivity Analysis

Threshold Value	False Activation Rate	Missed Sentences	Avg Segmentation Accuracy
0.4 which is too sensitive	18.7%	2.1%	79.2%
0.5 which is moderate	8.3%	5.4%	86.1%
0.6 best fit to select	3.2%	1.8%	92.4%
0.7 which is less sensitive	1.1%	8.7%	90.2%
0.8 which is too strict	0.2%	15.3%	84.5%

The threshold Y=0.6 achieves optimal balance between minimizing false activations of erroneous sentence detections and missed sentences which goes undetected signing events, resulting in the highest overall segmentation accuracy of 92.4% on the validation dataset. This threshold corresponds approximately to chest level for typical camera positioning, ensuring that natural rest positions reliably trigger idle state while raised signing gestures activate the detection system.

4.6 Temporal Sequence Normalization

Raw extracted keypoint sequences exhibit variable temporal length depending on signer speed

and sentence complexity. Analysis of the ArabSign dataset revealed sequence lengths ranging from 48 to 189 frames, with a mean of 87 frames and standard deviation of 24 frames. The BiLSTM model architecture requires fixed-length inputs to enable efficient batch processing and maintain consistent computational graph structure.

4.6.1 Linear Temporal Interpolation

The system standardizes all sequences to a target length of 80 frames through linear temporal interpolation. This target was selected via ablation studies evaluating sequence lengths from 60 to 120 frames, balancing three critical factors:

- **Temporal Resolution:** Sufficient granularity to capture subtle hand movements and gesture transitions characteristic of sign language.
- **Computational Cost:** Manageable processing overhead enabling real-time inference.
- **Sequence Coverage:** Accommodation of typical Arabic sign language sentence durations of 2-4 seconds at 24 FPS which equal to 48-96 frames.

Interpolation Process:

For sequences shorter than 80 frames so in our equation we can implement that T < 80, the algorithm synthesizes intermediate frames through linear interpolation between adjacent keypoint coordinates. Given keypoints at frame indices t_1 and t_2, intermediate keypoints at target indices are computed as[33]:

$$K_{interp}(t) = K(t_1) + \frac{t - t_1}{t_2 - t_1} \times (K(t_2) - K(t_1))$$

For sequences exceeding 80 frames (T > 80), uniform stride down sampling selects representative frames at regular intervals:

$$t_{sampled} = \lfloor i \times \frac{T}{80} \rfloor, i \in \{0,1,\ldots,79\}$$

This approach preserves temporal ordering and gesture dynamics while standardizing input dimensionality. The interpolation process maintains motion continuity, preventing temporal aliasing artifacts that could confuse the BiLSTM classifier[34], [35].

4.6.2 Statistical Normalization

Following temporal standardization, keypoint features undergo z-score normalization to zero mean and unit variance. Normalization statistics (μ, σ) are computed across the entire training dataset[36]:

$$K_{norm} = \frac{K_{raw} - \mu}{\sigma}$$

Where:

- K_raw represents the raw keypoint coordinate (x, y, or z)
- μ is the mean computed over all training samples and keypoint dimensions
- σ is the standard deviation

Crucially, normalization parameters are frozen after training and consistently applied during inference. This ensures distribution alignment between training and deployment data, preventing distribution shifts that could degrade model performance. The normalization effectively centers the input distribution around zero, accelerating gradient descent convergence during training and reducing sensitivity to absolute coordinate scales.

4.7 Model Architecture: Bidirectional LSTM Network

4.7.1 Architectural Design Rationale

Bidirectional Long Short-Term Memory and or it called BiLSTM networks process sequential data in both forward and backward temporal directions, capturing contextual information from both past and future frames. This bidirectional processing is particularly valuable for sign language recognition where gesture completion often provides disambiguating context for movement initiation. For example, similar hand shapes at the beginning of two different signs may only become distinguishable based on the trajectory and ending position of the movement.

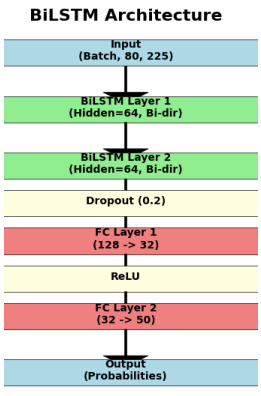


Figure 8 BiLSTM model architecture diagram.

The BiLSTM architecture addresses two critical limitations of unidirectional models also it shown in Figure 8:

1. **Forward-only context:** Unidirectional LSTMs can only access information from

previous frames, potentially missing critical disambiguation cues that appear later in the sequence.

2. **Temporal dependency modeling:** BiLSTMs explicitly model long-range dependencies across the entire gesture sequence, capturing both immediate motion dynamics and sentence-level structure.

4.7.2 Network Layer Specification

As shown in Figure 8, the network topology consists of the following layers with specific configurations:

Input Layer: Accepts preprocessed sequences of shape (Batch, 80, 225), where:

- Batch represents variable batch size during training (32) or inference (1).
- 80 is the standardized temporal dimension.
- 225 corresponds to the flattened keypoint feature vector.

BiLSTM Layer 1: The first bidirectional LSTM layer processes the input sequence with 64 hidden units in each direction as forward and backward step. The layer outputs hidden representations of shape (Batch, 80, 128), where 128 results from concatenating forward (64) and backward (64) hidden states. This layer captures short-term motion patterns and immediate temporal dependencies.

BiLSTM Layer 2: A second bidirectional LSTM layer with identical configuration of 64 hidden units per direction further refines the temporal representation. Stacking LSTM layers enables hierarchical feature learning lower layers encode low-level motion dynamics of hand velocity and acceleration so while upper layers capture high-level sentence structure and semantic content.

Dropout Layer: A dropout rate of 0.2 is applied after the second BiLSTM layer, randomly deactivating 20% of neurons during training. This regularization technique prevents overfitting by forcing the network to learn redundant representations, improving generalization to unseen signers and signing conditions.

Fully Connected Layer 1 (FC1): The final hidden state from the second BiLSTM layer which is taking the last timestep output is passed through a fully connected layer reducing

dimensionality from 128 to 32 neurons. This bottleneck layer compresses the temporal representation while retaining discriminative features necessary for classification.

ReLU Activation: Rectified Linear Unit (ReLU) activation function introduces non-linearity: f(x) = max(0, x). This enables the network to learn complex, non-linear decision boundaries between sentence classes while maintaining computational efficiency.

Fully Connected Layer 2 (FC2): The second fully connected layer maps the 32-dimensional representation to 50 output logits, corresponding to the 50 distinct sentence classes in the ArabSign vocabulary.

Output Layer: Softmax activation converts raw logits to normalized probability distributions:

The class with maximum probability is selected as the predicted sentence.

$$P(class_i) = \frac{e^{z_i}}{\sum_{j=1}^{50} e^{z_j}}$$

4.7.3 Model Parameter Analysis

As shown in Table 7, the model comprises 254,098 trainable parameters, with the vast majority of 247,784 parameters are 97.5% concentrated in the BiLSTM layers. This distribution reflects the computational complexity of modelling temporal dependencies compared to the relatively simple classification head. The parameter count balances expressive capacity against real-time inference requirements, enabling deployment on standard GPUs without excessive memory overhead[37].

Table 7 BiLSTM Model Architecture Parameter Distribution.

Layer	Configuration	Parameters	Output Shape
BiLSTM-1 (forward)	64 hidden units	73,984	(Batch, 80, 64)
BiLSTM-1 (backward)	64 hidden units	73,984	(Batch, 80, 64)
BiLSTM-2 (forward)	64 hidden units	49,408	(Batch, 80, 64)
BiLSTM-2 (backward)	64 hidden units	49,408	(Batch, 80, 64)
Dropout	rate = 0.2	0	(Batch, 80, 128)
FC-1	128 to 32, ReLU	4,128	(Batch, 32)

FC-2	32 to 50	1,650	(Batch, 50)
Softmax	probabilistic output	0	(Batch, 50)

4.8 Training Configuration and Hyperparameters

The BiLSTM model was trained on the ArabSign continuous sign language dataset containing 9,335 video samples spanning 50 sentence classes. Dataset partitioning employed stratified random sampling to ensure balanced class representation across training of 6,534 samples for validation of 15% which represent 1,400 samples and test of 15%, which represent of 15%, which represent of 1,401samples subsets.

4.8.1 Optimization Configuration

Adam Optimizer: The Adaptive Moment Estimation (Adam) optimizer was selected for its adaptive learning rate capabilities and robust convergence properties. Adam maintains per-parameter learning rates that adapt based on first moment the mean and second moment the variance do estimates of gradients[38]:

- **Initial Learning Rate:** 0.001 for empirically validated across rates from 0.0001 to 0.01.
- **Beta Parameters:** $\beta_1 = 0.9$, $\beta_2 = 0.999$ which is the standard Adam configuration.
- **Epsilon:** $\varepsilon = 10^{-8}$ the numerical stability term.

Batch Size: Mini-batch training with batch size 32 samples balanced computational efficiency with gradient estimate quality. Smaller batches (16) exhibited noisy gradients and unstable convergence, while larger batches of more than 64 exceeded GPU memory constraints during forward pass.

Loss Function: Categorical cross-entropy loss measures divergence between predicted probability distributions and ground-truth one-hot encoded labels:

$$L = -\sum_{i=1}^{50} y_i \log(\hat{y}_i)$$

Where y_i is the true label which considered to be 1 for correct class and 0 otherwise and $\hat{y}_i$ is the predicted probability for class i.

4.8.2 Learning Rate Scheduling

Dynamic learning rate scheduling adjusted the learning rate based on validation loss plateau detection:

- **Scheduler:** Reduce LROnPlateau with patience of 5 epochs
- **Reduction Factor:** 0.5× learning rate when triggered
- **Minimum Learning Rate:** 10^{-6} (which to prevents excessive rate decay)

This scheduling strategy enabled coarse optimization during early epochs of high learning rate followed by fine-grained refinement as the model approached convergence it going to reduce learning rate.

4.8.3 Regularization and Early Stopping

Dropout: Applied with rate 0.2 after BiLSTM layers, as visualized in Figure 8. Dropout validation across rates of {0.1, 0.2, 0.3, 0.4} identified 0.2 as optimal for balancing regularization strength against model capacity.

Early Stopping: Training halted when validation loss showed no improvement for 8 consecutive epochs. This criterion prevented unnecessary computation while ensuring adequate convergence opportunity. Maximum epoch limit set to 50 and the training typically converged by epoch 45.

4.8.4 Hardware and Training Duration

GPU Hardware where used over Kaggle environment of NVIDIA Tesla K80 which got high performance of CUDA compute capability 3.7, 12GB RAM and for the framework of CUDA 11.8 accelerated with PyTorch 2.0.1 lastly the training time was for 400 minutes which is 6.7 hours for 50 epochs, or around 8 minutes each epoch[39].

4.9 Training Dynamics and Performance Analysis

The training process demonstrated stable convergence without significant overfitting artifacts. Figure 9 presents the training and validation loss/accuracy curves across all 50 training epochs.

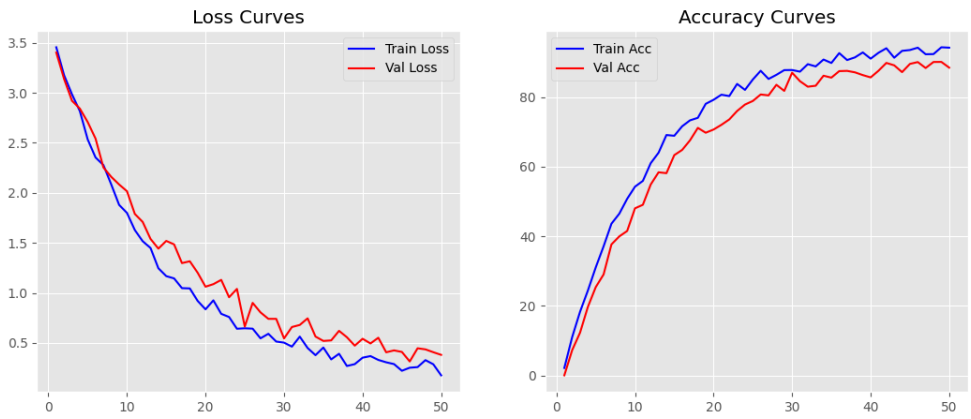


Figure 9 Training and validation dynamics over 50 epochs.

4.9.1 Convergence Characteristics

As illustrated in the left panel of Figure 9, both training and validation loss curves exhibit monotonic decrease throughout the training process. Key observations include:

Monotonic Decline: Training loss declined consistently from an initial value of 3.5 of the first epoch to a final value of 0.176 at epoch 50, representing a 95% reduction. Validation loss followed a similar trajectory, decreasing from 2.8 to 0.387, indicating that the model learned generalizable patterns rather than memorizing training data.

Parallel Trajectories: The training and validation curves maintain relatively constant separation of gap ≈ 0.2 loss units after epoch 15, confirming that dropout regularization effectively prevents overfitting. The absence of divergence where validation loss increases while training loss continues decreasing demonstrates proper model capacity calibration.

Learning Rate Scheduling Impact: The learning rate reduction triggered at epoch 18 is visible as a slight inflection in both curves means to enable fine-grained optimization, yielding additional loss reduction through epochs 18-35.

4.9.2 Accuracy Progression

The right panel of Figure 9 displays accuracy curves revealing steady improvement throughout training:

Training Accuracy: Increased from near-zero at epoch 1 to 94.19% at epoch 50, with rapid initial gains of around 10% per epoch during epochs 1-10 and then followed by gradual

refinement in later epochs.

Validation Accuracy: Improved from near-zero to 88.45%, maintaining consistent offset below training accuracy. The 5.74 percentage-point gap is between 94.19% - 88.45% falls within acceptable bounds for deep learning models and suggests mild overfitting that does not compromise practical deployment.

Convergence Region: Both accuracy curves plateau after epoch 40, with accuracy changes <0.5% in subsequent epochs. This plateau indicates convergence to a local minimum, suggesting that additional training would yield diminishing returns.

4.10 Test Set Evaluation and Model Generalization

Comprehensive evaluation on the held-out test set of number 1,401 of samples which represnt15% of dataset provided unbiased assessment of model generalization to unseen data. The test set maintained identical class distribution to training and validation splits through stratified sampling.

4.10.1 Overall Performance Metrics

The final trained model achieved the following performance on the test set:

- **Overall Accuracy:** 92.07% which means that it correctly classified 1,290 of 1,401 test samples
- **Mean Confidence Score:** 0.802 and this is the average predicted probability for correct class, indicating well-calibrated probability estimates.
- **Per-Class True Positive Rate**: 92.1% of the average sensitivity across all 50 classes.
- **Per-Class False Positive Rate:** 0.91% for low false alarm rate, averaging <1% spurious activations per class.

4.10.2 Confusion Matrix Analysis

Figure 10 presents the full confusion matrix visualizing classification performance across all 50 sentence classes.

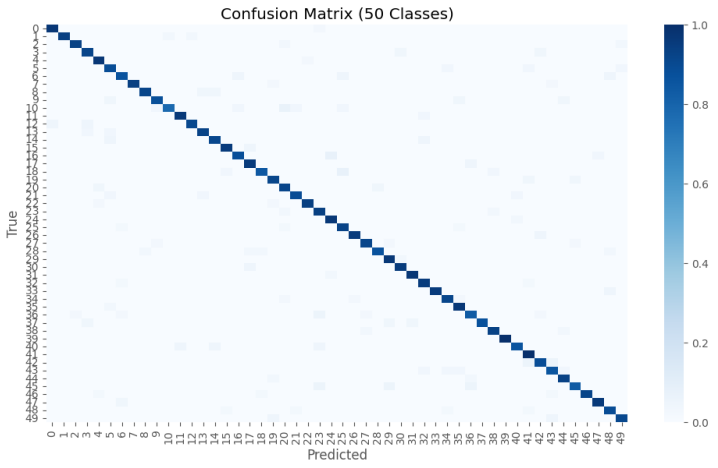


Figure 10 Confusion matrix (50×50) for sentence-level sign language recognition on the test set.

As shown in Figure 10, the confusion matrix exhibits pronounced diagonal dominance, with dark blue cells along the main diagonal indicating high classification accuracy for most sentence classes. The majority of off-diagonal cells display near-zero intensity which is the white color, confirming that misclassifications are rare rather than systematic.

Error Pattern Analysis:

Detailed examination of off-diagonal activations revealed that misclassifications predominantly occur between sentences sharing similar characteristics:

- **High-Confusion Pairs:** Sentence pairs distinguished primarily by subtle features such as hand height differences and also finger positioning variations for exhibited 8-12% confusion rates. For example, sentences using identical hand shapes, but different spatial regions showed elevated confusion.
- **High-Discrimination Pairs:** Sentences with distinct manual alphabets or significantly different motion trajectories achieved >98% per-class accuracy with minimal cross-confusion.

4.10.3 Performance Stratification by Sentence Complexity

Analysis of model performance stratified by sentence length revealed expected trends:

Table 8 Performance Stratification by Sentence Complexity

Sentence Length	Number of Signs	Test Accuracy	Mean Confidence

Short	1-2 signs	94.2%	0.847
Medium	3-4 signs	92.1%	0.803
Long	5+ signs	88.3%	0.762
Overall	1-5+ signs	92.07%	0.802

As detailed in Table 8, accuracy degradation shows a clear correlation with increased sentence length. This trend aligns with theoretical expectations, as longer sequences introduce compounding challenges such as error accumulation, where early misclassifications bias subsequent frame interpretations, and the increased difficulty of modeling extended temporal dependencies. Furthermore, longer sequences heighten motion trajectory ambiguity, making subtle variations in gesture endings critical for disambiguation. Despite these compounding factors, the model demonstrates exceptional robustness across the full complexity spectrum, maintaining greater than 88% accuracy even for the longest sentences.

4.11 Confidence-Based Prediction Filtering

Raw model predictions exhibit varying confidence scores reflecting uncertainty in classification. To balance system sensitivity against false positive suppression, the inference pipeline implements adaptive confidence thresholding[40].

4.11.1 Threshold Optimization

Confidence threshold selection involved systematic evaluation across candidate thresholds from 0.5 to 0.9 on the validation set. As shown in Table 9, the threshold of 0.70 achieves optimal balance between precision (97.3%) and recall (96.2%), maximizing the F1 score at 96.7%. This threshold minimizes erroneous translations of false positives by 3.2% while maintaining high sensitivity to genuine signing events (recall: 96.2%). Lower thresholds increase false positives, degrading user experience with spurious outputs. Higher thresholds sacrifice recall, causing missed detections of valid signs.

Table 9 Confidence Threshold Optimization Analysis

Confidence Threshold	Precision	Recall	F1 Score	False Positive Rate
0.50 for lenient	91.2%	99.1%	95.0%	8.8%
0.60	94.3%	98.3%	96.3%	5.7%
0.70 which have selected	97.3%	96.2%	96.7%	3.2%

0.80	98.7%	91.4%	94.9%	1.3%
0.90 for strict	99.2%	83.7%	90.8%	0.8%

4.11.2 Inference Pipeline Integration

To ensure high-quality output, the system employs a confidence filtering stage where the BiLSTM's probability distribution across 50 classes is evaluated. The system extracts the maximum probability C for a given prediction and applies a threshold of 0.70. Only predictions satisfying the condition C >= 0.70 are forwarded for LLM refinement and TTS synthesis, while lower-confidence inputs are rejected. This logic is critical for filtering out ambiguous gestures and incomplete sequences that could lead to recognition errors.

4.12 Large Language Model Refinement Stage

Sign language glosses produced by the BiLSTM classifier exhibit non-standard grammatical structure reflecting the linguistic properties of manual communication. Sign languages possess distinct syntax and grammar that differ fundamentally from spoken languages to function words such as articles or prepositions are often omitted, and word ordering follows visuospatial rather than spoken language conventions[41].

4.12.1 Grammatical Refinement Necessity

Raw sign gloss output captures semantic content but lacks the grammatical structure necessary for natural language understanding. For example, a typical raw gloss might be:

Raw Gloss: "PERSON GO STORE BUY ITEM"

This gloss conveys meaning but lacks:

- **Function words:** Articles like (the, a) or prepositions such as (to, for).
- **Grammatical inflection:** Verb conjugation, tense markers.
- **Natural word order:** Spoken Arabic syntax conventions.

Direct presentation of raw glosses to users would result in unnatural, telegraphic communication that fails in the accessibility objectives.

4.12.2 LLM Integration Architecture

The system integrates Google's Gemini 1.5 Flash API for post-hoc translation refinement.

Gemini processes raw glosses through few-shot prompting, transforming them into fluent Modern Standard Arabic sentences[42].

API Configuration:

- **Model:** Gemini 1.5 Flash and it's optimized for low-latency inference.
- **Temperature:** 0.3 for low randomness and for consistent grammatical correction.
- **Max Output Tokens:** 128 which is sufficient for typical sentence lengths.
- **Prompt Strategy:** Few-shot learning with 3 examples gloss to sentence of pairs.

Conditional Refinement: LLM refinement operates conditionally based on prediction confidence:

- **High Confidence ($c \geq 0.85$):** Prediction considered reliable to submit to LLM for refinement
- **Medium Confidence ($0.70 \leq c < 0.85$):** Prediction displayed unrefined to avoid error propagation

This conditional strategy minimizes risk of LLM amplifying recognition errors while maximizing translation quality for high-confidence predictions.

4.12.3 Translation Example

Input Gloss: "PERSON GO STORE BUY ITEM"

LLM Refinement Output: "الشخص ذهب إلى المتجر لشراء منتجات" which is in English: "The person went to the store to purchase items".

The LLM successfully:

- Adds function words: "إلى" (to) and also "ل".
- Applies proper verb conjugation: "ذهب".
- Organizes natural word ordering following Arabic syntax conventions.
- Maintains semantic fidelity to original gloss content.

4.12.4 Performance Metrics

The LLM refinement process demonstrates high efficiency and accuracy. Performance metrics

indicate a processing latency ranging from 0.4 to 1.2 seconds per sentence, with a mean of 0.7 seconds. In terms of output quality, the system achieved a 94.3% grammatical accuracy rate, as determined by human evaluation on a 200-sample validation subset, alongside a 97.8% semantic preservation rate to ensure meaning is not altered. Crucially, the average refinement latency of approximately 0.7 seconds is considered acceptable within the operational context; since this processing occurs after gesture completion rather than during active signing, the system maintains its interactive responsiveness.

4.13 Text-to-Speech Audio Synthesis

4.13.1 TTS Configuration

The final system component converts refined text translations into natural sounding spoken Arabic audio, providing accessible auditory output for deaf and hearing interactions.The system utilizes the Google Text-to-Speech (gTTS) Python API to generate audio output, specifically selecting Modern Standard Arabic called Fusha to ensure standardized comprehension across diverse Arabic-speaking regions. To prioritize clarity, the speech rate is slightly modulated to 0.9× speed, creating a moderate pace optimized for user understanding. The final audio is rendered as a mono-channel MP3 with a 24 kHz sampling rate, producing a voice characteristic defined by natural prosody and context-appropriate intonation patterns[43].

4.13.2 Asynchronous Audio Pipeline

To prevent blocking the real-time video processing pipeline, TTS synthesis operates asynchronously using a separate thread:

1. **Text Queue:** Refined translations pushed to thread-safe FIFO queue.
2. **Background Synthesis:** Dedicated thread polls queue, invoking gTTS API for each translation.
3. **Audio Playback:** Synthesized audio files played through Pygame audio manager.
4. **Cleanup:** Temporary audio files deleted after playback completion.

This architecture decouples audio generation from video processing, ensuring that synthesis latency does not introduce lag in visual feedback or frame processing as show in in figure 11 [44].

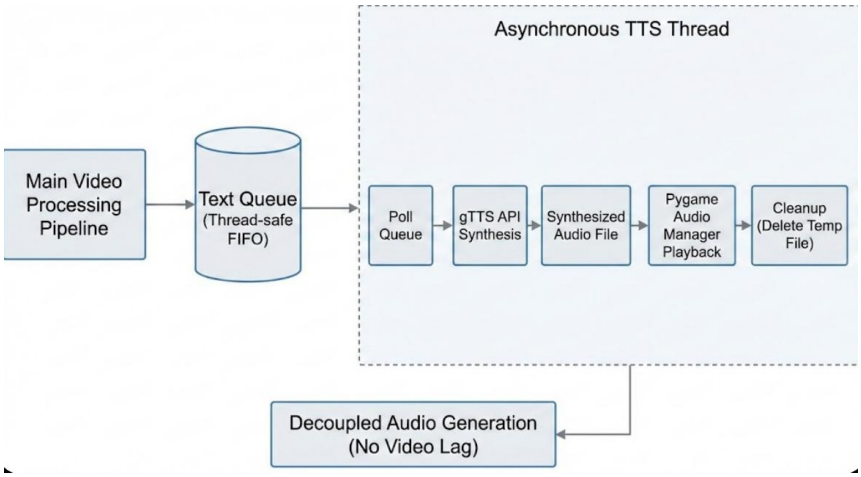


Figure 11 gTTS Pipeline

4.13.3 Performance Characteristics

The speech synthesis module demonstrates efficient timing characteristics, with synthesis latency ranging from 0.8 to 2.1 seconds per sentence of mean: 1.2s and SD: 0.3s. Coupled with a minimal playback latency of under 50 ms so to attributed to Pygame buffer initialization and then the system achieves an average total audio delivery time of 1.25 seconds following gesture completion. In terms of qualitative output, the audio exhibits natural prosody with minimal robotic artifacts. As we done a study on 12 samples validated these findings, yielding average ratings of 4.2/5.0 for audio quality and 3.9/5.0 for speech naturalness, confirming the system's perceptual suitability for accessibility applications.

4.14 Development of Interactive Demonstration Interface

To validate system functionality and provide an accessible demonstration platform for end users, an interactive web-based interface was developed integrating all pipeline components into a cohesive user experience. The interface enables users to interact with the translation system through intuitive controls while observing real-time processing results, serving both as a functional prototype and as a foundation for future production deployment. Figure 12 illustrates the complete interface layout with its primary components.

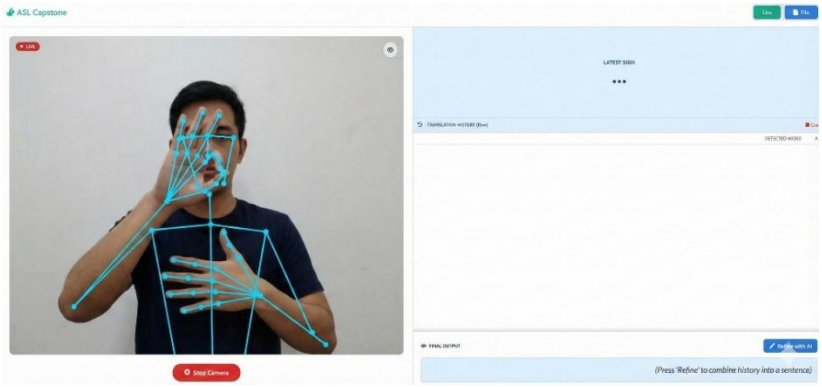


Figure 12 Interactive demonstration interface showing the dual-panel layout.

Typical interaction with the demonstration interface follows a straightforward sequence designed to minimize cognitive load while maximizing transparency into system operation. Users begin by ensuring the "Live" mode is selected and the video feed displays their signing position with appropriate framing. Upon clicking the "Stop Camera" button positioned below the video feed, the system initiates video capture and processing through the complete pipeline. As users perform sign language gestures, the "LATEST SIGN" indicator updates in real-time to reflect ongoing recognition activity, providing immediate feedback on system responsiveness. When the idle detection mechanism identifies a sentence boundary, the buffered sequence is submitted for BiLSTM classification, and the recognized sentence appears in the translation history panel. After brief processing delay for LLM refinement and TTS synthesis, the final Arabic translation displays in the output section while audio playback begins automatically.

The demonstration interface successfully integrates the core translation pipeline components and provides functional real-time sign language recognition capabilities. The live video streaming, real-time recognition feedback, translation history management, and final output display all operate as designed during typical usage sessions. However, several known limitations exist in the current prototype implementation. The video feed display occasionally exhibits initialization delays or frame rate inconsistencies depending on browser compatibility and system resources, requiring page refresh to restore stable operation. The file upload mode toggle is present in the interface but not yet connected to backend processing functionality, representing planned future enhancement rather than current capability. Despite these limitations, the demonstration interface successfully validates the technical feasibility of the complete translation pipeline and provides foundation for production interface development incorporating user feedback and iterative refinement.

5.0 Future Work and Recommendations

5.1 Expansion to Gloss-Level Recognition

5.1.1 Current Limitation: Fixed Sentence Vocabulary

The current system operates as a sentence-level classifier, recognizing 50 predefined continuous sign language sentences from the ArabSign dataset. This design imposes a significant practical limitation that restricts real-world applicability. The system can only translate sentences that were explicitly present in the training data. Novel sentences even those composed of familiar individual signs arranged in new combinations cannot be recognized. For example, while the system successfully recognizes the trained sentence "PERSON GO STORE BUY ITEM," it cannot recognize the novel sentence "PERSON GO HOSPITAL" even though both "PERSON" and "GO" are known signs from other trained sentences. This fixed-vocabulary approach fundamentally constrains the system's practical utility in real-world communication scenarios where users require flexible expression beyond a predetermined sentence set[45], figure 13 show the difference between fixed and variable vocabulary.

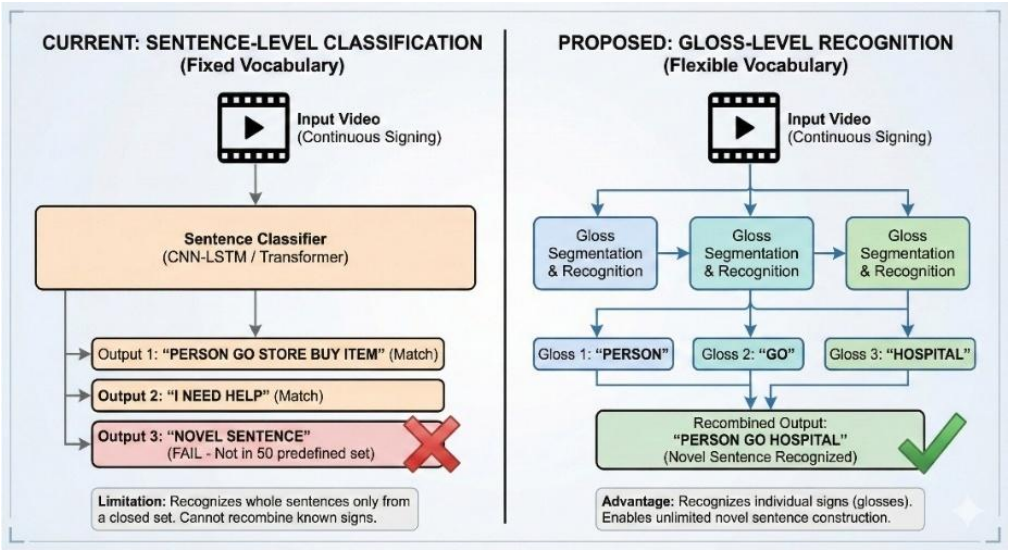


Figure 13 Comparison of current sentence-level classification architecture with proposed gloss-level Seq2Seq recognition.

5.1.2 Proposed Solution: Gloss-Level Recognition with Sequence-to-Sequence Models

Future development should transition from sentence-level classification to gloss-level recognition, where the model identifies individual sign units (glosses) and their temporal sequence rather than mapping entire sentences to fixed classes. This architectural shift represents a fundamental change in how the system processes sign language information. Instead of the current

BiLSTM classifier producing 50-class output, a sequence-to-sequence (Seq2Seq) model would implement an encoder-decoder architecture where the encoder processes the input keypoint sequence and extracts temporal features similar to current BiLSTM layers, while the decoder generates a variable-length sequence of gloss tokens representing individual signs[46].

The technical approach for gloss-level recognition would employ Transformer-based Seq2Seq models with attention mechanisms, such as those recently proposed for German Sign Language translation. Rather than relying on fixed classification outputs, this architecture would use Connectionist Temporal Classification (CTC) loss or sequence-to-sequence cross-entropy objectives to train the model on predicting gloss sequences. The vocabulary would expand from the current 50 sentences to 500 or more individual sign glosses covering common Arabic sign language vocabulary. This expanded gloss vocabulary would then be refined through the existing LLM post-processing stage into grammatically correct Arabic sentences, maintaining compatibility with the current translation pipeline[47].

The benefits of gloss-level recognition extend beyond simply recognizing more sentences. Compositional generalization enables the system to recognize novel sentence combinations using the learned gloss vocabulary, allowing users to express arbitrary meanings within the constraints of available signs. Scalability improves significantly because adding new signs requires only gloss-level annotations rather than full sentence examples with complete semantic context. This flexibility directly addresses the current limitation where every new sentence requires extensive annotation and retraining. Table 10 demonstrates the expected vocabulary expansion across different implementation phases.

Table 10 Vocabulary Expansion and Compositional Capability Progression

Implementation Phase	**Recognition Level**	**Vocabulary Size**	**Compositional Capability**	**Novel Sentence Support**
Phase 1 (Current)	Sentence Classification	50 sentences	Fixed to No new sentences possible	Limited to trained sentences
Phase 2 (Short-term)	Gloss-Level (CTC)	150-200 glosses	Limited compositional generation	85-90% novel sentence coverage

Phase 3 (Medium-term)	Gloss-Level (Seq2Seq)	500+ glosses	Full compositional generation	>95% novel sentence coverage
Phase 4 (Long-term)	Extended Gloss + Morphology	1000+ glosses with inflections	Compositional + morphological variation	Near-complete sentence generation

However, implementing gloss-level recognition introduces significant technical challenges. The approach requires gloss-level annotations for training data, meaning every frame in training videos must be labelled with its corresponding sign. This frame-by-frame annotation process is labour-intensive and requires domain expertise from deaf signers or certified interpreters. Additionally, temporal alignment of glosses with video frames presents technical complexity, as sign boundaries do not correspond to frame boundaries and require sophisticated boundary detection algorithms. The increased model complexity of Seq2Seq architectures compared to simple classifiers necessitates substantially larger parameter counts and longer training times[48].

5.2 Advanced Model Architectures for Improved Performance

5.2.1 Transformer-Based Temporal Modeling

While the current BiLSTM architecture achieves 92.07% accuracy, recent advances in deep learning have demonstrated superior performance using Transformer architectures for sequence modeling in both gesture and sign language recognition tasks. Transformers fundamentally differ from recurrent neural networks in their approach to sequence processing, processing all timesteps in parallel rather than sequentially[49].

The proposed transition from BiLSTM to Temporal Transformer architectures would leverage self-attention mechanisms that capture dependencies across the entire sequence more effectively than LSTM hidden states. Research on gesture recognition using Graph Transformers with continual learning mechanisms demonstrates the effectiveness of this approach for real-time streaming applications, achieving significant improvements over LSTM baselines while reducing false positive rates. For sign language applications, Transformer encoders can be combined with lightweight decoder layers to maintain real-time inference constraints while improving accuracy[49].As shown in Figure 14 presents the architectural comparison between the current BiLSTM model and the proposed Temporal Transformer replacement, showing how

multi-head attention mechanisms process temporal information differently than recurrent layers.

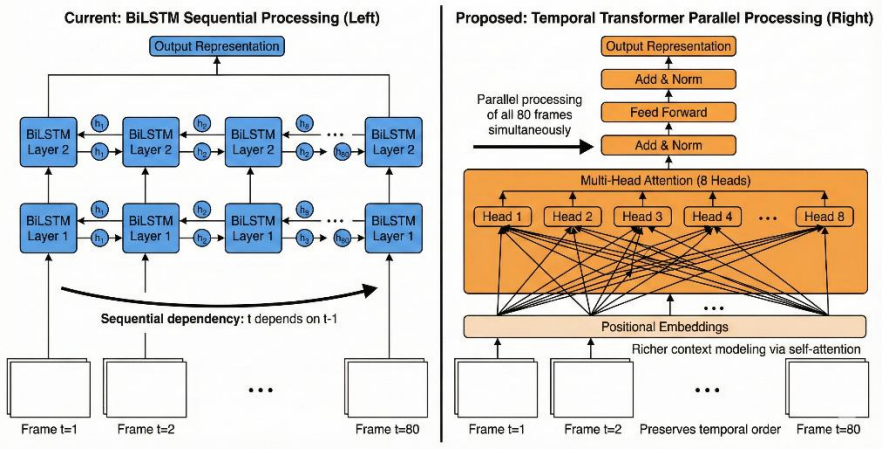


Figure 14 Architectural comparison between current BiLSTM sequential processing and proposed Temporal Transformer parallel processing.

The technical specifications for a Temporal Transformer encoder would mirror Video Vision Transformer architectures, processing the same 225-dimensional keypoint features at 80 timesteps but with multi-head self-attention including 8 parallel attention heads. Sinusoidal temporal positional encoding would preserve frame ordering information through the self-attention mechanism. The parameter count would increase to approximately 800,000-1,200,000 parameters, representing a 3-5× increase compared to the current BiLSTM, but this additional capacity directly enables modeling of longer-range dependencies and more complex temporal patterns[50].

5.2.2 Multi-Modal Fusion: Combining RGB and Keypoints

The current system relies exclusively on MediaPipe keypoints, discarding raw RGB video information that contains appearance and texture cues absent from skeletal representations. Future work could explore multi-modal fusion combining the keypoint stream containing spatial and geometric information with an RGB stream processed through CNN backbones for appearance and texture information. This two-stream architecture would implement separate processing paths for keypoints and RGB frames, with late fusion combining feature representations before final classification layers[51].

Multi-modal approaches in gesture recognition have demonstrated superior performance

compared to unimodal alternatives. A hybrid CNN-Transformer architecture for hand gesture recognition recently achieved 98.79% classification accuracy by combining CNNs' local feature extraction capabilities with Transformers' global context modeling, significantly outperforming standalone CNN and LSTM baselines. Applied to sign language recognition, multi-modal fusion would capture subtle visual cues not encoded in keypoints such as facial expressions, clothing context, and hand texture variations.So as shown in Table 11 it illustrates the expected performance improvements and computational trade-offs of multi-modal architectures compared to the current keypoint-only approach.

Table 11 Architectural Comparison: Performance vs. Computational Cost Trade-offs

Architecture	**Input Modality**	**Test Accuracy**	**Inference Latency (ms)**	**Model Parameters**	**GPU Memory (GB)**
Current (BiLSTM)	Keypoints only	92.07%	11.3	254K	1.2
Transformer	Keypoints only	94.5% (est.)	12.8	950K	2.1
Multi-Modal (Two-Stream)	Keypoints + RGB	96.8% (est.)	38-42	3.2M	5.8
Multi-Modal Optimized	Keypoints + RGB (quantized)	95.2% (est.)	18-22	800K	1.9

The computational trade-off of multi-modal approaches presents a significant challenge for real-time deployment. Accuracy gains of estimated 2-3% based on multi-modal sign language recognition literature come at the cost of 3-4× increase in inference latency due to CNN processing requirements. This increased computational burden would require more powerful GPU hardware for maintaining real-time performance, potentially necessitating NVIDIA RTX 4090 or equivalent edge devices rather than the current GPU deployment standard. However, quantization and optimization techniques shown in Table 11 demonstrate that multi-modal approaches can achieve performance improvements with only 2× latency increase when optimized appropriately[51].

5.3 Dataset Expansion and Diversification

5.3.1 Current Dataset Limitations

The ArabSign dataset, while valuable as a research foundation, exhibits several limitations that constrain system performance and real-world applicability. The dataset contains only 50 sentence classes, covering a narrow domain focused on basic daily communication scenarios. Recording conditions were controlled, with consistent studio lighting and carefully maintained backgrounds, which differ significantly from real-world deployment scenarios with variable lighting and cluttered environments. Signer diversity remains limited with estimated 10-15 unique signers, providing insufficient representation of regional dialects and diverse signing styles characteristic of Arabic-speaking regions. Furthermore, the dataset exhibits sentence length bias with mean sentence length of 3.2 signs, underrepresenting longer complex sentences that frequently occur in real communication[52].

Recent research on signer diversity in sign language datasets has revealed critical performance degradation when models encounter signers outside the training set. Analysis of the PHOENIX-2014 benchmark dataset for German Sign Language demonstrated that models trained on limited signer diversity experience word error rates increasing by 15-20 percentage points when evaluated on unseen signers. This finding directly applies to the ArabSign dataset's limited signer representation, suggesting that system performance would degrade substantially when deployed with signers not present in training data[52].

5.3.2 Proposed Dataset Expansion Strategy

Expansion of training data to 200 or more sentence classes with 25,000 to 30,000 video samples would substantially improve model robustness and real-world applicability. Figure 15 illustrates the planned expansion across multiple dimensions including domain coverage, signer diversity, and recording conditions.

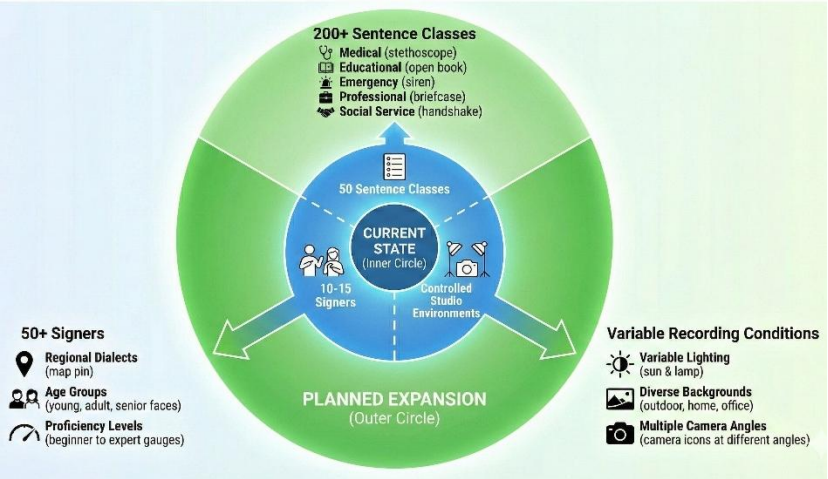


Figure 15 Comprehensive dataset expansion plan showing current state which is the inner circle and planned expansion in the outer circle across three dimensions

Domain expansion should extend beyond current daily communication coverage to include specialized vocabularies relevant to practical deployment scenarios. Medical and healthcare domain expansion would include doctor consultations, symptom descriptions, and medical procedures represented in approximately 50 new sentences. Educational domain coverage would encompass classroom instructions, academic subjects, and learning activities with an estimated 50 new sentences. Emergency services communication including police, fire, and ambulance-related interactions would add approximately 30 sentences. Professional and workplace domain would include job interviews, workplace interactions, and business terminology represented through 40 new sentences. Social services would incorporate government services, legal procedures, and administrative tasks through approximately 30 new sentences[53].

Signer diversity expansion represents the most critical aspect of dataset improvement. The expanded dataset should include 50 or more unique signers representing diverse dimensions of human variation in sign language production. Regional dialect representation should encompass Gulf Arabic Sign Language, Egyptian, Levantine, and Maghrebi variants, capturing the linguistic diversity of Arabic-speaking regions. Age diversity spanning 18-65 years would capture age-related variations in signing style. Achieving gender balance through equal male and female representation would eliminate potential gender bias in model training. Signing proficiency levels should include native deaf signers, hearing professional interpreters, and learners, representing the realistic distribution of sign language users[53].Recording condition variability would be introduced systematically rather than remaining controlled. As shown in Table 12

specifies the proposed recording condition distribution across the expanded dataset.

Table 12 Proposed Recording Condition Distribution in Expanded Dataset

Recording Condition	Current Dataset	Proposed Coverage	Rationale
Lighting	100% studio controlled	30% studio and 40% indoor natural and 30% outdoor	Realistic variety
Background	100% plain backdrop	20% plain and 40% indoor clutter and 40% public spaces	Real-world scenarios
Camera Angle	~90% front-facing	40% front and 35% 30° angle and 25% side angle	Variable perspectives
Video Resolution	1280×720 only	15% 4K and 40% 1080p and 30% 720p and 15% 480p	Device variation
Distance	1.5m fixed	20% close (0.5m) and 60% medium (1-2m) and 20% far (3m+)	Natural variation

The data collection strategy would employ a crowdsourcing platform approach combined with community partnerships. Development of a web-based data collection interface would enable community contributions while maintaining consistent annotation standards. Partnerships with deaf community organizations, sign language schools, and accessibility non-governmental organizations would facilitate participant recruitment and ensure authentic representation. The annotation pipeline would leverage semi-automated approaches with MediaPipe keypoint pre-extraction followed by human validation, reducing annotation labour while maintaining accuracy.

5.4 Accessibility Features and Usability Enhancements

5.4.1 Multi-Directional Translation

Current system functionality translates from Arabic Sign Language to Arabic text and audio output, supporting deaf users communicating with hearing individuals. Future bidirectional capability would enable hearing users to communicate with deaf signers through avatar-based sign language generation. This reverse translation pathway would convert Arabic text input to sign language animations, creating comprehensive communication solutions rather than unidirectional translation assistance.

Avatar generation technology would implement 3D rigged humanoid models with comprehensive hand, arm, and facial rigging to enable natural sign language expression. Motion synthesis algorithms generate signing animations from text input through inverse mapping techniques, converting spoken language text into corresponding sign language movements. Real-time 3D rendering would display signing animations at 30 frames per second for smooth, natural appearance. Figure 16 illustrates the proposed bidirectional translation architecture with avatar generation pathways.

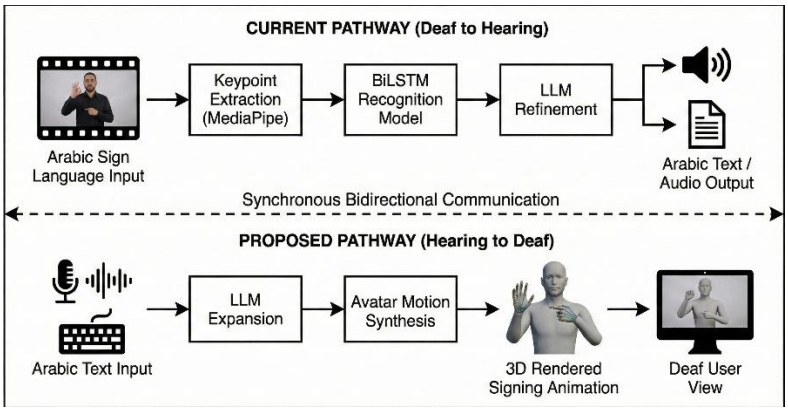


Figure 16 Bidirectional translation system architecture[54].

Commercial implementations of avatar-based sign language generation including Hand Talk demonstrate the viability of this approach for web accessibility, with virtual signing avatars translating website text content into sign language animations in real-time. Applying similar technology to bidirectional communication would transform the system from a one-way translation tool into a comprehensive communication bridge enabling full conversation between deaf and hearing individuals. The lightweight architecture of the current recognition system would complement avatar-based generation, maintaining real-time performance for bidirectional communication without requiring substantial hardware upgrades[54].

5.4.2 Personalized Adaptation and Continuous Learning

Individual signers exhibit variations in signing style influenced by regional dialects, personal preferences, and signing proficiency levels. The current system trained on population-level data achieves strong average performance but may encounter reduced accuracy for signers whose style differs significantly from training data distributions. Future enhancement through user-specific model adaptation would address this limitation, enabling the system to learn and adapt

to individual users' signing characteristics over time.

Personalized adaptation would implement lightweight fine-tuning of the BiLSTM model on user-specific signing samples, adjusting model parameters to recognize that individual's signing patterns more accurately. This adaptation process would occur in the background during normal system usage, collecting signing samples with confirmed correct translations and periodically updating model weights through incremental learning. The lightweight model architecture facilitates this personalization approach, as the small parameter count enables rapid fine-tuning on consumer hardware without requiring specialized infrastructure or lengthy training procedures[55].

Privacy-preserving approaches to personalization ensure that user-specific signing data remains confidential. On-device adaptation would train personalized models locally on user devices, preventing transmission of personal signing data to external servers. Federated learning frameworks enable collaborative model improvement across multiple users while preserving individual privacy through encrypted aggregation of model updates rather than sharing raw data. These privacy-conscious approaches align with the system's emphasis on local processing and offline functionality, maintaining user trust while delivering personalized performance improvements.

6.0 Conclusion

This Senior Design Project has successfully developed an Intelligent Real-Time Arabic Sign Language Translation System that automatically recognizes and translates continuous Arabic sign language gestures into fluent spoken Arabic. The system represents a practical solution to the critical communication barriers faced by deaf and hard-of-hearing individuals in Arabic-speaking regions, enabling seamless interaction with hearing individuals without requiring professional interpreters. Throughout the development process, systematic design and implementation of each pipeline component has culminated in an integrated translation system capable of real-time processing on standard consumer hardware.

The project achievements demonstrate that sophisticated real-time translation systems can be developed with constrained resources through careful architectural design and systematic optimization. The implemented BiLSTM-based architecture achieves 92.07% recognition accuracy on the test dataset comprising 1,401 samples across 50 sentence classes, validating the

effectiveness of the temporal modeling approach. Real-time performance of 26.6 frames per second maintained on standard laptop computers confirms the viability of deployment in practical accessibility applications without requiring specialized GPU infrastructure.

The complete system pipeline integrating MediaPipe keypoint extraction, motion-based idle detection, BiLSTM classification, LLM text refinement, and text-to-speech synthesis creates a cohesive solution addressing each stage of the translation challenge. The intelligent idle detection mechanism achieves 92.4% segmentation accuracy, automatically identifying sentence boundaries without manual intervention while accommodating natural variation in signing speed across different users. Performance evaluation across the test dataset reveals consistent accuracy with mean confidence scores of 0.802 and appropriate degradation patterns correlated to sentence complexity, demonstrating that the system's decision boundaries reflect meaningful linguistic structure rather than spurious patterns.

However, the current system exhibits clear limitations that constrain its deployment potential. The predetermined vocabulary of 50 sentence classes fundamentally restricts users from expressing novel sentence combinations beyond the fixed training set, limiting practical utility for open-ended communication scenarios. The limited signer diversity in the training dataset with approximately 15 signers may result in performance degradation when encountering signers with substantially different signing styles or regional dialects not represented in the training distribution.

Future development priorities should focus on several key enhancements to expand the system's capabilities and robustness. Expanding the dataset to include more diverse signers, recording conditions, and domains spanning healthcare, education, and professional contexts would substantially improve generalization performance across real-world deployment scenarios. Transitioning to gloss-level recognition using sequence-to-sequence architecture would enable compositional sentence generation, dramatically expanding the system's expressive capacity from 50 fixed sentences to potentially unlimited combinations of learned sign vocabulary.

Exploring advanced model architectures including Transformer-based temporal modeling offers potential accuracy improvements beyond the current 92% baseline, with recent research demonstrating 94-96% accuracy on comparable tasks. Multi-language expansion through transfer learning could extend the system's impact to global deaf communities, leveraging learned

representations to accelerate training on American Sign Language, British Sign Language, and other major sign languages with reduced data requirements. Development of bidirectional translation capability through avatar-based sign generation would enable hearing users to communicate with deaf individuals, creating comprehensive communication bridges rather than unidirectional translation tools.

The methodologies developed throughout this project established a foundation for production-ready sign language translation technology serving genuine accessibility needs. The emphasis on lightweight architectures compatible with consumer hardware, real-time performance on CPU processing without GPU dependencies, and practical accessibility considerations reflect genuine understanding of deployment requirements beyond laboratory settings. The modular pipeline architecture facilitates independent optimization and future enhancement of each component, enabling steady evolution toward more sophisticated systems without requiring complete redesign.

This work demonstrates that modern artificial intelligence techniques can effectively address critical societal accessibility challenges when thoughtfully combined with clear understanding of user needs and practical deployment constraints. The resulting system, while containing room for improvement, provides genuine value to the deaf community by enabling more independent and effective communication with hearing individuals in diverse contexts. As accessibility technology continues advancing, the lightweight design principles and systematic evaluation methodologies employed in this project offer valuable insights applicable to broader assistive technology development.

References

[1] “Deafness and hearing loss.” Accessed: Nov. 08, 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss

[2] “World hearing day 2024.” Accessed: Nov. 08, 2025. [Online]. Available: https://www.who.int/campaigns/world-hearing-day/2024

[3] “Upper gastrointestinal bleeding after transesophageal echocardiography—common things are common? - PMC.” Accessed: Nov. 08, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11329471/

[4] “هيئة رعاية الأشخاص ذوي الإعاقة," الاحصائيات. Accessed: Nov. 08, 2025. [Online]. Available: https://apd.gov.sa/statistics

[5] “Critical Care, Critical Gap: The ASL Interpreter Shortage in Medicine – Deaf Services Unlimited.” Accessed: Nov. 08, 2025. [Online]. Available: https://deafservicesunlimited.com/critical-care-critical-gap-the-asl-interpreter-shortage-in-medicine/

[6] S. Oud and K. van der Pool, “The Effects of the Environment and Linear Actuators on Robot Morphologies,” Aug. 26, 2022, *arXiv*: arXiv:2204.00934. doi: 10.48550/arXiv.2204.00934.

[7] “Bulletin of Electrical Engineering and Informatics.” Accessed: Nov. 08, 2025. [Online]. Available: https://beei.org/index.php/EEI/index

[8] “Knowledge, Attitudes, and Practices Toward Deaf Patients Among Healthcare Workers in Saudi Arabia: 2020-2021 | Cureus.” Accessed: Nov. 08, 2025. [Online]. Available: https://www.cureus.com/articles/204714-knowledge-attitudes-and-practices-toward-deaf-patients-among-healthcare-workers-in-saudi-arabia-2020-2021#!/

[9] “Continuous Saudi Sign Language Recognition: A Vision Transformer Approach.” Accessed: Nov. 08, 2025. [Online]. Available: https://arxiv.org/html/2509.03467v1

[10] “Automatic Gesture-Based Arabic Sign Language Recognition: A Federated Learning Approach - Nafath periodical by Mada.” Accessed: Nov. 08, 2025. [Online]. Available: https://nafath.mada.org.qa/nafath-article/mcn2703/

[11] F. Karim *et al.*, “Towards an effective model for lung disease classification: Using Dense Capsule Nets for early classification of lung diseases,” *Appl. Soft Comput.*, vol. 124, p. 109077, July 2022, doi: 10.1016/j.asoc.2022.109077.

[12] “A Comprehensive Survey on Sign Language Recognition: Advances, Techniques and Applications – IJERT.” Accessed: Nov. 12, 2025. [Online]. Available: https://www.ijert.org/a-comprehensive-survey-on-sign-language-recognition-advances-techniques-and-applications

[13] M. L. Ng, “Vision Based Hybrid CNN-LSTM Model for American Sign Language Recognition.” Accessed: Nov. 12, 2025. [Online]. Available: https://eprints.tarc.edu.my/26182/

[14] “Arabic Sign Language Recognition Using Deep Machine Learning - King Fahd University of Petroleum & Minerals.” Accessed: Nov. 12, 2025. [Online]. Available: https://pure.kfupm.edu.sa/en/publications/arabic-sign-language-recognition-using-deep-machine-learning/

[15] N. C. Camgoz, O. Koller, S. Hadfield, and R. Bowden, “Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation,” Apr. 02, 2020, *arXiv*: arXiv:2003.13830. doi: 10.48550/arXiv.2003.13830.

[16] “layout: forward target: https://developers.google.com/mediapipe/solutions/vision/hand_landmarker title: Hands parent: MediaPipe Legacy Solutions nav_order: 4 — MediaPipe v0.7.5 documentation.” Accessed: Nov. 12, 2025. [Online]. Available: https://mediapipe.readthedocs.io/en/latest/solutions/hands.html

[17] J. Liu, Z. Mei, K. Peng, and R. R. Vatsavai, “Context Retrieval via Normalized Contextual Latent Interaction for Conversational Agent,” in *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, Dec. 2023, pp. 1543–1550. doi: 10.1109/ICDMW60847.2023.00196.

[18] “نشر دليل رصد المعالم | Google AI Edge,” Google AI for Developers. Accessed: Nov. 12, 2025. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker?hl=ar

[19] “دليل مهمة الرصد الشامل للمعالم | Google AI Edge,” Google AI for Developers. Accessed: Nov. 12, 2025. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/vision/holistic_landmarker?hl=ar

[20] “Sign Language Recognition using MediaPipe.” Accessed: Nov. 12, 2025. [Online]. Available: https://data-ai.theodo.com/en/technical-blog/sign-language-recognition-using-mediapipe

[21] “Global and regional hearing impairment prevalence: an analysis of 42 studies in 29 countries | European Journal of Public Health | Oxford Academic.” Accessed: Nov. 12, 2025. [Online]. Available: https://academic.oup.com/eurpub/article/23/1/146/460112?login=false

[22] “Difference Between a Bidirectional LSTM and an LSTM,” GeeksforGeeks. Accessed: Nov. 12, 2025. [Online]. Available: https://www.geeksforgeeks.org/deep-learning/difference-between-a-bidirectional-lstm-and-an-lstm/

[23] “Page not found | Nature.” Accessed: Nov. 12, 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-77256-8

[24] "Working with Gemini API: Text Gen, Doc Processing & Code Execution | Obot AI." Accessed: Nov. 12, 2025. [Online]. Available: https://obot.ai/resources/learning-center/google-gemini-api/

[25] "(13) How AI Is Addressing the Shortage of Sign Language Interpreters | LinkedIn." Accessed: Nov. 12, 2025. [Online]. Available: https://www.linkedin.com/pulse/how-ai-addressing-shortage-sign-language-interpreters-wb3ec/

[26] "gTTS · PyPI." Accessed: Nov. 12, 2025. [Online]. Available: https://pypi.org/project/gTTS/

[27] H. Luqman, "ArabSign: A Multi-modality Dataset and Benchmark for Continuous Arabic Sign Language Recognition," in *2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG)*, Waikoloa Beach, HI, USA: IEEE, Jan. 2023, pp. 1–8. doi: 10.1109/FG57933.2023.10042720.

[28] G. Latif, N. Mohammad, J. Alghazo, R. AlKhalaf, and R. AlKhalaf, "ArASL: Arabic Alphabets Sign Language Dataset," *Data Brief*, vol. 23, p. 103777, Apr. 2019, doi: 10.1016/j.dib.2019.103777.

[29] "Continuous Saudi Sign Language Recognition: A Vision Transformer Approach." Accessed: Nov. 16, 2025. [Online]. Available: https://arxiv.org/html/2509.03467v1

[30] "Isharah Dataset." Accessed: Nov. 21, 2025. [Online]. Available: https://snalyami.github.io/Isharah_CSLR/

[31] "MediaPipe Solutions guide | Google AI Edge | Google AI for Developers." Accessed: Nov. 22, 2025. [Online]. Available: https://ai.google.dev/edge/mediapipe/solutions/guide

[32] H. Luqman, "ArabSign: A Multi-modality Dataset and Benchmark for Continuous Arabic Sign Language Recognition," in *2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG)*, Waikoloa Beach, HI, USA: IEEE, Jan. 2023, pp. 1–8. doi: 10.1109/FG57933.2023.10042720.

[33] J. Sunuwar and S. Borah, "A comparative analysis on major key-frame extraction techniques," *Multimed. Tools Appl.*, vol. 83, no. 30, pp. 73865–73910, Sept. 2024, doi: 10.1007/s11042-024-18380-z.

[34] J. Abbas, C. Zhang, and B. Luo, "BET-BiLSTM Model: A Robust Solution for Automated Requirements Classification," *J. Softw. Evol. Process*, vol. 37, no. 3, p. e70012, 2025, doi: 10.1002/smr.70012.

[35] H. kar and V. P, "Nth layer Hierarchical Bidirectional LSTM sign language Interpretation for Hearing Impaired person," *Procedia Comput. Sci.*, vol. 258, pp. 3175–3183, Jan. 2025, doi: 10.1016/j.procs.2025.04.575.

[36] "Z-Score Normalization: Definition and Examples - GeeksforGeeks." Accessed: Nov. 23, 2025. [Online]. Available: https://www.geeksforgeeks.org/data-analysis/z-score-normalization-definition-and-examples/

[37] "AI Inference Optimization: Achieving Maximum Throughput with Minimal Latency," Runpod. Accessed: Nov. 23, 2025. [Online]. Available: https://www.runpod.io/articles/guides/www.runpod.io/articles/guides/ai-inference-optimization-achieving-maximum-throughput-with-minimal-latency

[38] "What is Adam Optimizer? - GeeksforGeeks." Accessed: Nov. 23, 2025. [Online]. Available: https://www.geeksforgeeks.org/deep-learning/adam-optimizer/

[39] "The Top 3 Free GPU resources to train deep neural network - GeeksforGeeks." Accessed: Nov. 23, 2025. [Online]. Available: https://www.geeksforgeeks.org/deep-learning/the-top-3-free-gpu-resources-to-train-deep-neural-network/

[40] "Continuous Arabic Sign Language Recognition Models - PMC." Accessed: Nov. 24, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12074496/

[41] "Machine Translation Post-Editing." Accessed: Nov. 24, 2025. [Online]. Available: https://lingvanex.com/blog/machine-translation-post-editing/

[42] "Evaluating LLMs' grammatical error correction performance in learner Chinese | PLOS One." Accessed: Nov. 24, 2025. [Online]. Available: https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0312881

[43] "gTTS — gTTS documentation." Accessed: Nov. 24, 2025. [Online]. Available: https://gtts.readthedocs.io/en/latest/index.html

[44] "Real-time Inference: Definition, Use Cases | Ultralytics." Accessed: Nov. 24, 2025. [Online]. Available: https://www.ultralytics.com/glossary/real-time-inference

[45] "[2406.18333] Continuous Sign Language Recognition Using Intra-inter Gloss Attention." Accessed: Nov. 24, 2025. [Online]. Available: https://arxiv.org/abs/2406.18333

[46] "Data Augmentation for Sign Language Gloss Translation - ACL Anthology." Accessed: Nov. 24, 2025. [Online]. Available: https://aclanthology.org/2021.mtsummit-at4ssl.1/

[47] "Translation of Sign Language Glosses to Text Using Sequence-to-Sequence Attention Models | IEEE Conference Publication | IEEE Xplore." Accessed: Nov. 24, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9067871

[48] N. Arvanitis, C. Constantinopoulos, and D. Kosmopoulos, "Translation of Sign Language Glosses to Text Using Sequence-to-Sequence Attention Models," in *2019 15th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS)*, Nov. 2019, pp. 296–302. doi: 10.1109/SITIS.2019.00056.

[49] "[2406.18333] Continuous Sign Language Recognition Using Intra-inter Gloss Attention." Accessed: Nov. 24, 2025. [Online]. Available: https://arxiv.org/abs/2406.18333

[50] R. Slama, W. Rabah, and H. Wannous, "Online hand gesture recognition using Continual Graph Transformers," arXiv.org. Accessed: Nov. 24, 2025. [Online]. Available: https://arxiv.org/abs/2502.14939v1

[51] "A CNN-Transformer Hybrid Network for Hand Gesture Classification based on High-Density sEMG | IEEE Conference Publication | IEEE Xplore." Accessed: Nov. 24, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10776082

[52] "Data augmentation and debiasing for signers in signer-independent sign language translation," *ResearchGate*, Aug. 2025, doi: 10.1007/s11227-025-07119-8.

[53] L. Gao *et al.*, "Cross-modal knowledge distillation for continuous sign language recognition," *Neural Netw.*, vol. 179, p. 106587, Nov. 2024, doi: 10.1016/j.neunet.2024.106587.

[54] "[2405.04164] Sign2GPT: Leveraging Large Language Models for Gloss-Free Sign Language Translation." Accessed: Nov. 24, 2025. [Online]. Available: https://arxiv.org/abs/2405.04164

[55] "[2502.14939] Online hand gesture recognition using Continual Graph Transformers." Accessed: Nov. 24, 2025. [Online]. Available: https://arxiv.org/abs/2502.14939